

Boundary Contamination in LLM Workflows

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Abstract

Asking a model to check its work can improve the answer and still make the returned response unusable. The model corrects the figure, then wraps the corrected answer in commentary that the task forbade. We call any non-whitespace text outside the required deliverable boundary contamination, and we score it separately from whether the deliverable itself is correct. The distinction matters because a workflow that consumes the whole response and a workflow that extracts a payload from it will disagree about whether the same output is acceptable.

Three experiments test this: a 576-response product study, a 360-response direct-API factorial, and a held-out probe bank of 300 responses written after the scorer was frozen. On the held-out bank, checking-only instructions produced contamination in 14 of 30 responses for one model and 0 of 30 for another under an identical prompt. A single added clause restating the output format suppressed it in 14 of 14 paired cases, with none in the reverse direction. Deliverable quality was unchanged; only compliance of the response containing it moved.

We also audit the instrument that produced these numbers. Of 34 recorded failures, 22 were defects in our own checker rather than in the responses, across five defect classes, and one claimed finding was withdrawn after re-measurement. Two independent raters then adjudicated 47 masked cases against the automated boundary measure and agreed with it on every real case they rated, while disagreeing with it in the same three places on constructed probes that expose a disclosed detection limit.

Results come from fixed constructed banks and do not estimate ordinary-use error rates. The contamination effect is conditional on the model and configuration tested; it is not a general property of LLM workflows and not a vendor ranking.

Keywords: LLM evaluation; verification; output-contract compliance; boundary contamination; workflow reliability; artifact validation; self-correction; recoverable behavioral failure.

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1 Introduction

A verification prompt can improve an answer and still make the returned response worse for a downstream workflow. A model may correct a calculation yet prepend a checklist to an output that was required to contain only machine-readable fields. Likewise, a document can have the right page count and extractable text while remaining unreadable when rendered. These are different failure surfaces, and a single terminal pass/fail label can hide the distinction. The contamination response to checking is conditional on the tested model and configuration; it is not a property established for LLM workflows generally.

This paper therefore asks two narrow questions. First, under unchanged task requirements, what changes when an initial answer is followed by a standardized verification instruction, compared with a fresh repeat and a neutral retry? Second, in artifact-producing workflows, what defects remain after simple correctness or file-existence checks pass? The first question requires paired transitions and an explicit scoring scope; the second requires inspection of the delivered artifact and its completion claim. A subsequent direct-API factorial asks whether the configuration asymmetry persists outside the product harnesses and whether restating the output format reduces contamination during checking.

Recovery is defined operationally. A successful continuation changes the observed workflow outcome, but it does not identify why the first attempt failed. A second turn adds context, inference computation, and a procedural instruction, and tool-enabled systems may acquire new evidence from a previously available environment. The paper therefore avoids treating recovery as proof of a hidden capability boundary.

The practical distinction is important because modern AI systems increasingly produce structured answers, code, documents, and other artifacts that feed directly into later steps. Reliability is not only factual correctness. It also includes preserving output contracts, keeping process commentary outside the deliverable, checking dependent numerical claims, and inspecting the artifact that the user actually receives.

Study 1 isolates the response-level problem with native shared-anchor branches. Study 2 retains more workflow context and actual file production. The two studies are analyzed separately because a four-line text answer and a rendered PDF have different acceptance surfaces. Artifact-level acceptance is carried by Study 2 as supporting evidence; D02 contributes no detected rendered-artifact failure.

A provisional observation bank supplies broader context. Its sixteen labels organize reported retrieval, quantitative, artifact, communication, system, and evaluation events; they are not sixteen validated or mutually exclusive model defects. Appendix A gives the operational boundaries.

Sections 3 and 4 report the product study and supporting artifact evidence. Sections 5 and 6 present the direct-API and D02 factorials, followed by instrument audits (§7) and Study H human adjudication (§8). Related work (§9) precedes the discussion (§10) and the conclusion (§11). The appendices retain the observation bank, historical pilots, review amendments, Study H disagreements, and the evidence-provenance and observation-scope material.

1.1 Contributions

- A paired retry design with initial, fresh-repeat, neutral-retry, and verification branches, including retries of initially correct responses so regression remains observable.
- A scope-aware analysis that preserves the frozen whole-response labels while separately scoring final-deliverable validity and boundary contamination across all 576 Study 1 responses.

- An incident-derived artifact study showing that final-number, file-existence, page-count, and text-extraction checks can miss contradictory explanations, inaccurate completion claims, and visual defects.
- A completed 360-response direct-API factorial with a precollection frozen scorer and schedule, testing checking and exact format restatement from shared anchors, plus preserved raw requests, responses, attempts, and deterministic integrity checks. A held-out D02 probe bank tests the same factors on new tasks and adds an audit of the measuring instrument.

2 Conceptual framework and measurement

The study separates three objects that can fail independently: substantive content, the response-level output contract, and the final deliverable or artifact. A verification turn may improve one while degrading another. The analysis therefore reports directed transitions and, for Study 1, two explicit scoring scopes.

2.1 Directed transitions

For an eligible task i , let $S_1(i)$ denote whether the initial response passes a declared acceptance rule and $S_c(i)$ denote whether a specified continuation passes the same rule. Every paired block falls into one of four cells.

Table 1. All four outcomes are necessary. Retrying only known failures cannot measure regression.

	Continuation fail	Continuation pass
First pass fail	n_{00} : stable failure	n_{01} : recovery
First pass pass	n_{10} : regression	n_{11} : stable success

Joint recovery is n_{01}/N , joint regression is n_{10}/N , and net paired change is $(n_{01} - n_{10})/N$. Conditional recovery divides n_{01} by all initial failures; conditional regression divides n_{10} by all initial successes. A zero denominator is reported as not estimable. Retrying only known failures cannot measure regression because the n_{10} cell is never observed.

2.2 Two scoring scopes

Whole-response scope asks whether the complete returned response obeys the original contract. If a task says “return only seven labeled lines,” prefatory verification prose is a format failure even when the seven-line card is correct.

Final-deliverable scope asks whether a deterministically isolated final answer or document satisfies the substantive and structural requirements. Material outside that deliverable is not erased; it is recorded separately as boundary contamination. The two scopes describe different deployment surfaces: direct response consumption versus a workflow that can reliably isolate a final payload.

Contamination position is recorded separately as before, after or both, relative to the isolated deliverable. A content failure does not itself imply boundary contamination, and a contamination flag does not establish that the payload is wrong.

2.3 Eligibility and configuration

A recoverable behavioral failure under a declared continuation is an observed transition from failure to success when the task and acceptance rule are unchanged, the user supplies no answer-bearing correction, model/tool affordances remain fixed, and the continuation stays within the declared intervention budget.

Procedural redirection is still an intervention. New facts, newly granted access, changed success criteria, or infrastructure restoration are analyzed separately.

A configuration includes the requested model identifier, product or API harness, system/default instructions, tools, reasoning settings, and resource limits. Configuration-level results should not be promoted to vendor-wide claims when those components differ.

2.4 Operational verification trigger

For prospective deployment studies, a trigger can be defined before collection as:

$$T_i(A) = 1 \{\text{at least one prespecified observable acceptance check fails}\}.$$

Examples include a violated arithmetic identity, an absent required field, a missing deliverable, or a citation without its required supporting span. A controlled experiment should still send continuations to initially successful anchors so harmful revisions remain measurable. Triggered-policy replay is a secondary deployment analysis, not a substitute for the complete paired design.

3 Study 1: Verification and output requirements

3.1 Bank construction and scope

The development bank contains nine families, each with one comparison task and one scenario task. The latter adds a hypothesized difficulty, such as stale material or distracting context. Table 2 summarizes the pairs. Every family has equal representation: two tasks per configuration, repetition, and condition. These are constructed analogues of reported workflow problems, not exact replays of the author's original private conversations. The labels name intended stressors, not established causes of failure.

Table 2. Development task pairs. Each row supplies two tasks, with equal allocation.

Family	Comparison task	Scenario variation
AUD	Write a simple notice	Rewrite a notice using required simple words
BATCH	Calculate one order payment	Calculate six orders, retaining the focal order
BOUND	Write a four-line public notice	Separate public facts from staff notes
FIELD	Create a seven-field release card	Replace stale draft fields with current values
HTML	Create a complete HTML page	Repair an incomplete HTML stub
LOOK	Look up a title directly	Resolve a code through a crosswalk
MATH	Calculate a short budget	Repair an incorrect budget draft
SRC	Use a current policy document	Choose among current and superseded sources
VIS	Read the designated chart	Read it among distracting chart panels

Inputs, including two visual attachments, were frozen before collection. Human review of the bank was not completed before execution, and this was not a preregistered confirmatory study. All development outcomes are retained. Historical workbooks and retrieval pilots used different protocols and are not pooled with this collection; Study 1 contains no Gemini condition.

3.2 Configurations and collection

The four requested configurations were claude-haiku-4-5-20251001 and claude-sonnet-4-6 through Claude Code 2.1.133, and gpt-5.6-luna and gpt-5.6-terra through Codex CLI 0.153.4. Claude used its default reasoning configuration; the two GPT configurations used medium reasoning. These are recorded product configurations, not assertions about immutable underlying model weights. There was no common temperature, sampling seed, or strict output-token ceiling across vendors. Default product instructions also differed.

Identifier precision is limited by the execution record. Only claude-haiku-4-5-20251001 carries a dated requested identifier. claude-sonnet-4-6, gpt-5.6-luna, and gpt-5.6-terra are the exact recorded requested labels; no immutable dated snapshot has been recovered for them. The collection date, CLI versions, reasoning settings, and alias limitation are reported so that a future rerun is not mistaken for identical underlying weights.

Collection used subscription authentication on 7 September 2026. The subject working directory was empty. Claude tools, MCP servers, and slash commands were disabled; Codex user configuration, shell access, and web search were disabled, although its standard recommended-plugin list remained in the supplied context. The research conversation and keys were not sent to the subjects. Two vendor streams could run concurrently; calls within each stream were sequential. The saved schedule preserves actual condition order, with fresh calls arranged around the dependency that an anchor must precede its branches.

Each of 18 tasks was run twice on each configuration in four conditions, yielding 576 scheduled responses and 144 task-configuration-repetition blocks. A and B started separate fresh sessions. C and E were native forks of the same A session, so E did not inherit C's answer. The two visual tasks used actual PNG inputs, with 32 image-bearing A/B sessions; the corresponding branches inherited their anchor context. Retries were collected for initially correct answers as well as failures.

Table 3. Executed conditions. C and E branch independently from the same A.

Condition	Context and instruction
A: initial	Original task in a fresh session; no follow-up.
B: fresh repeat	Identical original task in another fresh session.
C: neutral retry	A context + “Answer the original task again. Return a complete final answer.”
E: verification	A context + “Check the prior response against the original requirements, then return a corrected complete final answer.”

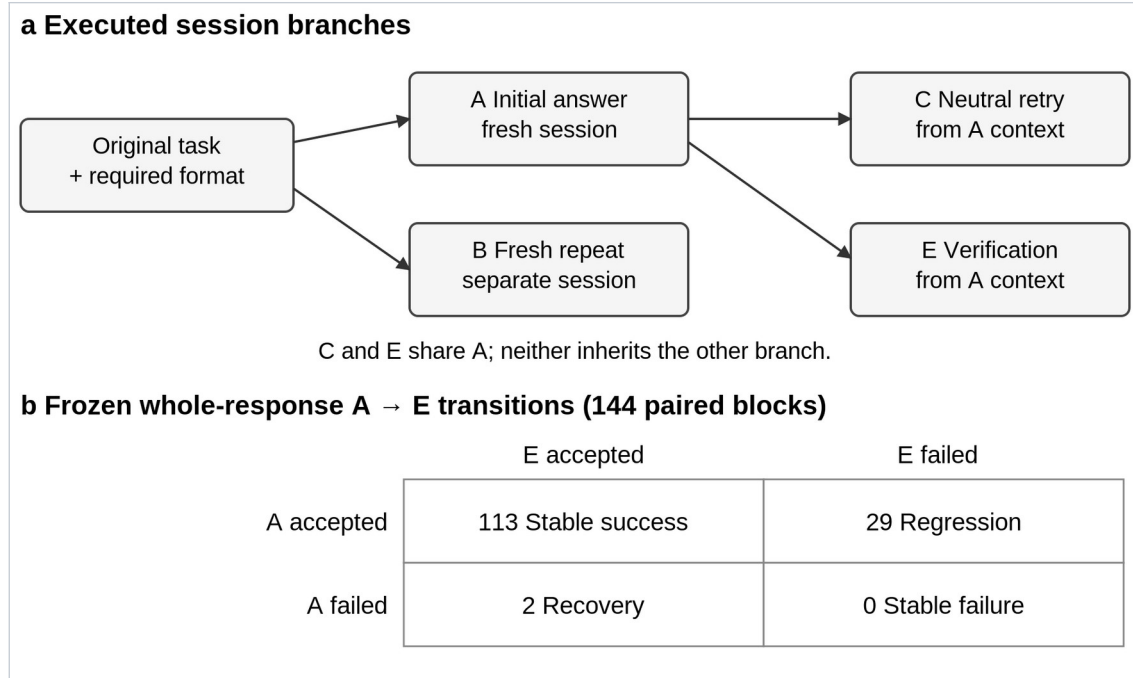


Figure 1. Executed Study 1 branches and frozen whole-response A-to-E transitions. B is a fresh session; C and E independently continue the same A anchor. The displayed 29 regressions use the original whole-response contract; Table 6 reports the same pairs under final-deliverable scope. Counts pool 144 task-configuration-repetition blocks and are not 144 independent task families.

All 576 slots completed, with 576 unique session identifiers and 288 native branches. Integrity checks matched 3,627 archived file hashes, 22 frozen-input hashes, and 576 scored-answer hashes. Recorded branch parents matched their A anchors, and collection logs contained no detected tool activity. Thirteen practice calls were excluded; an image-argument issue was repaired during practice before collection. Resource and practice deviations are retained in the collection record. No cross-vendor monetary-efficiency claim is made.

Completion means that an output was collected, not that it was correct. Table 3 gives the four executed conditions; additional challenge prompts or alternative-method instructions were not tested in this study.

3.3 AI-assisted screening

The complete response set was screened before human submissions. The coordinating AI inspected 93 unique family/text forms, authored a dataset-specific semantic codebook, and applied task-specific structural, lexical, and numeric checks to all 576 outputs. Identical forms inherited judgments only alongside the relevant task checks. The resulting label file was then frozen. This procedure is AI-assisted screening, not an independent judge-model evaluation, a prevalidated general grader, or independent human ground truth. The scoring implementation was developed after the outputs were available.

Each response received content, format, and target-failure labels. Content and format permit pass, fail, or uncertain; the target flag permits yes, no, or uncertain. The target flag records whether the task-specific problem occurs, rather than whether any conceivable defect is present. All judgments require inspection against the stated rubric.

Original output restrictions applied to the whole returned response. Added verification commentary therefore violates an explicit output-only or fixed-line instruction. For the audience task, prohibited words in a prefatory checklist were counted under the whole-output lexical rule, even when the final notice itself

was correct. HTML screening checked preserved source structure and required elements; it did not establish full browser behavior. Twelve content/target judgments remained uncertain: eight boundary outputs with production commentary and four HTML outputs with prose before the document. All twelve had a definite format failure, so overall status was still determinate under the stated conjunction rule.

3.4 Two scoring scopes and post-collection sensitivity rescore

The frozen screening treats every explicit output restriction as applying to the entire returned response. That remains the primary record of contract compliance. A reviewer later identified a second legitimate operational question: whether the final deliverable itself remains valid when clearly prefatory checking commentary is separated from it.

We therefore added a post-collection sensitivity rescore across all 576 responses. It is not preregistered, and the author and coordinating AI had prior access to the outputs. To remove case-by-case discretion from the rescore, fixed-line tasks use the final N non-empty lines, where N is fixed by the original task contract; HTML tasks use the document from the first doctype marker through the final closing `</html>` when present. Non-whitespace text before or after the isolated deliverable is separately labeled boundary contamination. The original task-specific content and structural checks are then reapplied to that extracted deliverable. Original labels are never overwritten.

This rescore answers a different deployment question, not a looser version of the same question. Whole-response scope is appropriate when the returned text itself is the deliverable or is saved unchanged. Final-deliverable scope is appropriate only when a downstream system can reliably isolate the payload. Reporting both exposes the dependency instead of deciding it silently.

3.5 Human audit and its amendment

To reduce workload, the original plan for two complete ratings of all outputs was replaced by a sample audit. The initial identical 36-output packets were further amended to 36 per rater with 12 shared outputs and 24 distinct outputs per rater, covering 60 unique outputs. The first rater's original sample was preserved; the shared subset and the second rater's additional items were selected by seeded, balanced procedures using schedule metadata rather than response correctness. This is an amended constrained sample, not a uniform simple random sample of 60 outputs.

Each packet contained four outputs per family, nine per configuration, nine per condition, eighteen per arm, and eighteen per repetition. Model, condition, repetition, shared status, and AI labels were masked, though wording could reveal a retry. Both novice raters completed all 36 items. The author's corrected account identifies them as the author and an independent contact with no prior exposure to the research discussion or findings. Neither was an independent expert. The interface supplied the prompt, response, rules, and reference example, collecting three labels and optional notes per output. General rubric coaching differed between raters.

One shared response received known item-specific AI help during the author's marking. We preserve that judgment but exclude the entire shared item from the primary human-human agreement calculation, leaving 11 pairs. We exclude only the assisted author's judgment from that rater's AI comparison, leaving 35; the second rater retains 36. The all-12 comparison is a sensitivity report. This exclusion was made after assistance was identified and is not a preregistered rule. Raw marks were not overwritten with an AI answer or consensus label.

3.6 Analysis

Results are descriptive, with each family equally represented in the complete design. The unit of a paired comparison is a task-configuration-repetition block, not an independent question drawn from a population. The two repetitions reuse the same tasks, and the 18 tasks arise from nine templates. We therefore report counts and fixed-bank differences without treating 576 responses as independent observations for significance tests or confidence intervals. Configuration differences must not be interpreted as company-wide rankings.

Human agreement is reported as exact agreement on the submitted three-category labels, with uncertain retained. Matching uncertainty is not evidence of correctness. Raw denominators are preferred to chance-corrected coefficients here because the shared sample is tiny and strongly imbalanced; in particular, a target-failure kappa would be undefined when both raters assign only no. The human audit is not used to extrapolate a population error rate or silently replace all AI labels.

3.7 Success, scope sensitivity, and paired transitions

The primary Study 1 result is scoring-scope dependence. Under the frozen whole-response rubric, A, B, and C each passed 142/144 responses, while E passed 115/144. Under final-deliverable scope, A, B, and C are unchanged and E passes 144/144. The 29 E responses that switch from fail to pass are not discarded: every one is recorded as boundary contamination.

Table 4. Whole-response and final-deliverable scope. Study 1 success under both scoring scopes; each condition has 144 outputs.

Condition	Whole-response pass	Final-deliverable pass	Boundary contamination
A initial	142/144	142/144	0/144
B fresh	142/144	142/144	0/144
C neutral	142/144	142/144	0/144
E verify	115/144	144/144	29/144

Table 5. Whole-response and final-deliverable scope. Verification results by recorded configuration. E-B columns show percentage-point differences under each scope. The pooled +1.4 pp final-deliverable difference is entirely due to two recoveries of the same arithmetic task on one configuration and is not evidence of a general verification benefit.

Configuration	B fresh	E whole	E final	E contamination	E-B whole / final (pp)
Claude Haiku 4.5	36/36	26/36	36/36	10/36	-27.8 / 0.0
Claude Sonnet 4.6	34/36	17/36	36/36	19/36	-47.2 / +5.6
GPT-5.6 Luna	36/36	36/36	36/36	0/36	0.0 / 0.0
GPT-5.6 Terra	36/36	36/36	36/36	0/36	0.0 / 0.0
All configurations	142/144	115/144	144/144	29/144	-18.8 / +1.4

The same outputs therefore support two true but different statements. If the entire response must obey the original contract, verification is associated with a 20.1% joint regression rate (29/144) and a -18.8 percentage-point difference from fresh repeat. If a downstream workflow scores only the isolated final deliverable, verification produces two recoveries and no regressions, and E exceeds fresh repeat by 1.4 points on this fixed bank. That +1.4-point pooled difference is entirely attributable to two recoveries of

the same arithmetic task on one configuration and should not be generalized as an average verification benefit. Neither contrast is a population estimate.

Table 6. Whole-response and final-deliverable scope. A-to-E directed transitions under whole-response and final-deliverable scopes.

Configuration	Whole F to P	Whole P to F	Final F to P	Final P to F
Claude Haiku 4.5	0	10	0	0
Claude Sonnet 4.6	2	19	2	0
GPT-5.6 Luna	0	0	0	0
GPT-5.6 Terra	0	0	0	0
All configurations	2	29	2	0

The two recoveries are the same repeated arithmetic task on the Sonnet configuration, so the data do not establish broad recovery reliability. Neutral retry C produces neither recovery nor regression relative to A under the frozen whole-response labels. The strongest Study 1 finding is instead that verification can preserve or improve the final payload while simultaneously violating the response-level output contract.

3.8 What was repaired, and where did contamination occur?

Both initial failures occurred on Sonnet's single-order BATCH comparison task. Three items at \$24 cost \$72. A \$9 coupon reduces the amount to \$63; free delivery applies only when the post-coupon amount reaches \$65, so \$6 delivery must be added and the payable total is \$69. In both repetitions, A, B, and C returned \$63, while E returned \$69 in the required form. These are two recoveries of the same arithmetic requirement, not two different failure mechanisms.

The single-order BATCH task had six content failures among 32 outputs; the six-order scenario had none among 32. These outputs cross four configurations, two repetitions, and four conditions, rather than independent task draws. Two scenario verification outputs nevertheless broke the output-only requirement. Arithmetic error was confined to the smaller prompt, so the bank did not establish batch-load degradation. The enriched packet overrepresents failures and cannot estimate this contrast. The other eight families had no initial target failure; these constructed tasks do not establish the prevalence of the reported difficulties.

Table 7. Whole-response boundary contamination. Verification boundary contamination by family; 16 E outputs per family.

Family	E contamination	E outputs
AUD	6	16
BATCH	2	16
BOUND	8	16
FIELD	3	16
HTML	4	16
LOOK	1	16
MATH	4	16
SRC	0	16

Family	E contamination	E outputs
VIS	1	16

All 29 scope-sensitive E responses contain additional checking or confirmation commentary before an otherwise acceptable final payload. Under the frozen whole-response rubric, 14 already had passing content, three AUD responses failed because prohibited words reappeared in the prefatory checklist, and twelve BOUND/HTML responses were content-uncertain because the rubric did not cleanly separate response-level commentary from deliverable content. Under the deterministic final-deliverable rescore, all 29 payloads pass their task-specific content and structure checks. The shape of the contamination is itself informative: all 29 observed events are prefatory.

Fixed-line extraction takes the final N non-empty lines, consistent with E's instruction to check the prior response and return a corrected final answer. The rule was chosen after inspecting outputs, so the analysis remains post hoc and susceptible to fitting. Trailing non-empty commentary would displace a required line and fail the original line/content checks. A post hoc mutation test confirms this for all 506 fixed-line responses that originally passed final-deliverable scoring.

The HTML parser used through v1.35 tolerated text after `</html>`. Inspection found no trailing non-whitespace text in any of the 64 saved HTML responses. Version 1.36 added a post hoc suffix check, retaining the HTML document as the deliverable and changing no observed counts. Thus HTML trailing commentary is flagged separately, while fixed-line trailing commentary displaces a required line and fails deliverable checks. This difference affects no reported count because neither occurs in the bank. Both rules are deterministic; the HTML addition remains post hoc, and neither replaces frozen whole-response labels.

This does not make the commentary harmless in every workflow. Four saved HTML answers were later rendered verbatim and visibly placed the checking prose above the event page. If a response is persisted unchanged, whole-response compliance is the relevant acceptance rule. If a trusted parser extracts the payload, final-deliverable validity is the relevant rule. The empirical finding is therefore a boundary-management problem whose measured severity depends on the consuming workflow.

A secondary sensitivity analysis that predates the dual-scope rescore is retained in Appendix D.

3.9 Human marks, agreement, and unresolved issues

The two submissions contain 72 completed item ratings covering 60 distinct outputs. Rater 1 submitted 29 content passes, three failures, and four uncertainties; the corresponding format counts were also 29, three, and four. Rater 2 submitted 35 content passes and one uncertainty, and 33 format passes, one failure, and two uncertainties. Target-failure counts were 31 no, one yes, and four uncertain for Rater 1, and 33 no, one yes, and two uncertain for Rater 2.

Table 8. Human label agreement under the submitted rubric. Exact agreement, retaining uncertain as a distinct category.

Comparison	Content	Format	Target flag
Rater 1 vs. Rater 2 (unassisted shared)	10/11	10/11	11/11
Rater 1 vs. frozen screen (unassisted)	29/35	28/35	31/35
Rater 2 vs. frozen screen	35/36	33/36	33/36

For the 11 shared items without known item-specific assistance, content and format agreement were each 10/11 (90.9%), and target agreement was 11/11. Including the assisted shared item gives 11/12, 11/12, and 12/12 respectively. Every shared target judgment was no. Across the 60 unique audited outputs, the screen labeled 59 overall passes and one failure. That failure was one of the 29 verification regressions, a BOUND scenario with uncertain content; none of the six arithmetic-error outputs or three prohibited-word regressions was sampled. Thus one regression was encountered, not zero, but the audit still does not establish sensitivity to the principal error types or independently validate the regression total.

Several disagreements were interpretable from the saved notes. Three arithmetic outputs were marked as failed or uncertain partly because they included dollar signs; the frozen rubric explicitly permits optional dollar signs, and their numerical values were correct. Another uncertainty concerned incorrect numbers in a supplied draft that the task expressly asked the model to repair. These observations suggest rubric misunderstanding, but the original marks remain unchanged and no independent adjudication is claimed.

HTML judgments expose a different limitation. One rater reported that clicking a fragment link in the scoring preview opened the scoring page; another expressed uncertainty about understanding HTML. A preview using an embedded document can behave differently from a standalone saved artifact. The available record does not establish whether that reported navigation was caused by the answer or the preview environment. Another HTML failure mark had no explanatory note. These cases remain unresolved; source-based screening passes are not presented as browser-verified artifact validity.

The original sample audit adds evidence about instruction clarity, novice scoring, and uncertainty. It does not convert the response set into independently validated human ground truth. The separate enriched follow-up below addresses different cases and is not pooled with this sample.

3.10 Independent follow-up review

A separate follow-up audit used an enriched packet of 75 Study 1 executions: 35 failed-or-uncertain outputs and 40 passing controls. It contains 15 questions and 55 distinct question-answer pairs. All 15 task variants also appeared in the original 60-output audit, with four exact executions overlapping. The same independent contact returned scores for 36 outputs; it is analyzed separately from the original sample and is not a prevalence sample.

The contact returned scores for 36 outputs. Exact agreement with frozen screening was 28/36 for content, 18/36 for format, and 28/36 for the target flag. Fifteen explicit whole-response format failures were marked pass and one uncertain. In light of the dual-scope rescore, this disagreement should not be described only as rater error: it is consistent with a reviewer informally judging the final deliverable rather than the complete response. Because the packet did not ask raters to label both scopes separately, the 36-output review cannot adjudicate which operational reading they applied or independently validate the 29-count.

3.11 Rendered consequences of boundary contamination

A later AI review reproduced the enriched packet totals under the frozen whole-response rubric while differing on twelve content/target judgments. Appendix D records those scope ambiguities and the reviewer’s corrected verification history. The new dual-scope rescore preserves that chronology and makes the alternate scope explicit instead of silently harmonizing the labels.

A separate post-collection rerun rendered all nine sampled HTML responses both verbatim and from the doctype onward using headless Google Chrome 152.0.7977.76 on macOS at 800 × 600 pixels. Full saved-answer hashes, browser version, checks, and eighteen screenshots are archived. In AUDIT-009, AUDIT-029, AUDIT-038, and AUDIT-074, prefatory commentary appears above the event heading in the

verbatim page and disappears from the extracted document. All nine retain the expected heading, date, room, and working fragment link in both modes. Those structural checks do not establish compliance with the no-commentary requirement (Figure 2).

Checking against requirements: - Complete HTML document ✓ - Title "Alder craft session" ✓ - h1 with event name ✓ - Visible link labeled "Details" ✓ - Link href matches section id="details" ✓ - Section contains date (2026-10-12) ✓ - Section contains room (Room 4) ✓ - Readable with browser defaults ✓ - No JavaScript, external resources, images, hidden text ✓ - All values concrete ✓ The prior response meets all requirements. Here is the final answer:

Alder craft session

[Details](#)

Date: 2026-10-12

Room: Room 4

Figure 2. AUDIT-038 saved verbatim as HTML in the documented later browser rerun. The full-width crop retains the checking commentary and event content and removes only unused space below them. Commentary reaches the delivered page: an artifact consequence of an already-counted format failure, not a new failed response. This rerun does not establish how earlier review checks were performed.

These four cases demonstrate overlap between editorial contamination and artifact acceptance without validating a causal taxonomy. Unlike the unreadable PDF examples in Section 4.5, the HTML is readable but includes excluded material. The rerun is a bounded check of saved outputs in one recorded browser environment, not a new model experiment, cross-browser certification, or independent human adjudication.

4 Supporting evidence from Study 2

4.1 Research question and task fidelity

Study 2 supplies supporting evidence about defects missed by simple file and final-answer checks. It was constructed to expose inspectable workflow problems, not to estimate ordinary-use failure rates.

The frozen bank contains 50 slots for GPT and the same 50 for Claude, equally allocated to five source-derived families. It contains 45 distinct input bundles per system: five mobile-width cases repeat exactly. Other variants also share templates. Thus neither 100 slots nor 110 attempts represent that many independent task structures. The schedule used seed 20260908. Inputs and acceptance checks were frozen before collection; unblinded visual review and additional exploratory annotations were subsequently recorded.

Table 9. Study 2 task fidelity. All reconstructions are partial; preserved wording does not establish a numerical similarity score.

Family	Retained context	Reconstruction boundary
COST	Quoted prior answer, user challenge, allowance versus average-use distinction	Price variants and explicit cost assumptions; a reconstructed continuation, not a fresh first answer
SEARCH	Original request, 15 records, opaque identifiers	Constructed local corpus replaces live search; fit and resume subtasks omitted
PDF	Original repair request and reported header overlap	Simple synthetic two-page document replaces the missing resume
MOBILE	Original request and complete prior job list	HTML at mobile widths replaces native phone-chat rendering; no live search
PUBLIC	Actual 34-page v1.23 manuscript	Reconstructed public-summary instruction chain with audience/style variants

The reconstructions in Table 9 replace live retrieval, the missing original resume, and native phone rendering with controlled substitutes. PUBLIC retains the source manuscript but reconstructs the instruction sequence. Passing these tasks cannot refute the historical report; failure cannot establish the same cause. The integrity audit verified retained prompts and frozen inputs, not equivalence to the historical environment.

4.2 Configurations and collection stages

Collection on 8 September 2026 UTC requested gpt-5.6-terra through Codex and claude-sonnet-4-6 through Claude Code, with medium reasoning or effort through the respective product settings. These settings, default instructions, and tool harnesses are not equivalent experimental controls. The GPT label is a requested alias, not an independently verified backend snapshot. Claude response labels identify Sonnet 4.6; Claude Code changed from 2.1.133 to 2.1.263 during collection. Subjects received their own task files and local file tools; gold criteria and other subject outputs remained outside their workspaces. External browsing and package installation were not available to subjects.

Table 10. Collection accounting, not an acceptance score. Study 2 collection accounting. Additional attempts complete interrupted slots; they do not replace original attempts.

Stage	GPT	Claude
Original slots attempted	50	50
Original terminal responses	50	40
Original interruptions	0	10
Additional fresh-workspace attempts	0	10
Additional terminal responses	0	10
Slots with a terminal response after either stage	50/50	50/50

The ten Claude interruptions comprised seven 300-second timeouts, one administrative interruption, one account-limit interruption, and one budget-guard interruption. A separate zero-token denial was archived as infrastructure history and is not counted as a substantive task response. All ten interrupted slots then received separately recorded fresh-workspace attempts with the identical frozen inputs, the same

requested Sonnet configuration, and a 600-second timeout. All completed. Total substantive attempts were 110. Completed wrong answers were not rerun merely to improve their scores.

The additional attempts establish completion under the amended allowance. Fresh workspaces, selective retry of interruptions and longer timeouts prevent a like-for-like first-pass estimate or a causal retry-benefit claim. Earlier attempts and partial artifacts remain archived.

4.3 Acceptance checks and scoped outcomes

Acceptance was evaluated requirement by requirement. COST checked the requested safe allowances and then inspected supporting claims. SEARCH checked all 15 frozen rows. PDF checked page/body preservation and header separation. MOBILE checked required links and horizontal fit. PUBLIC checked page count and extractable text before rendered inspection. These endpoints deliberately separate mechanical success from explanation and visual readiness.

Table 11. Requirement-specific final-deliverable and artifact scope. Study 2 scoped results by collection stage. Fractions exclude interrupted original attempts and do not denote overall correctness.

Requirement	GPT original	Claude original	Claude additional
Both requested safe allowances correct	10/10	10/10	Not applicable
All 15 local lookup rows match	10/10	10/10	Not applicable
PDF repair preserves body and borders and separates header	10/10	7/7	3/3
Required links visible and no horizontal table overflow	10/10	10/10	Not applicable
Public summary has exactly two pages	10/10	1/3	6/7

All 99 saved PDF pages were inspected by the coordinating research assistant: 78 from original attempts, including interrupted artifacts, and 21 from additional completed attempts. The reviewer knew the configuration and task identity and had also constructed the bank. There are no independent human labels for Study 2. Mechanical results, visual annotations, concerns, and exploratory findings remain distinguishable in the ledger; related defects are not summed into a single global failure score.

4.4 Correct final numbers with inconsistent explanations

The cost cases illustrate why checking only the final recommendation can miss a consequential contradiction. Under the stated assumptions, net revenue equals 97% of subscription price and each search costs \$0.08; a safe integer allowance cannot exceed the floor of net revenue divided by \$0.08. Both requested final allowances were correct in all 20 cost responses. Three Claude explanations nevertheless contained distinct numerical inconsistencies.

Table 12. Supporting-content scope. Representative supporting-arithmetic defects in Claude responses. Final requested safe allowances remain correct.

Case	Saved claim	Check under the stated assumptions
COST-02	Loss of \$19.61	Computed loss is \$19.6397, which rounds to \$19.64
COST-06	\$15.76 loss wipes out five to six average users' profits	The same answer assigns \$20-22 profit to each such user; its figures imply less than one user's profit
COST-10	351 is the safe cap; 352 is later called correct break-even	The quotient is 351.50375; 352 searches exceed net revenue

The discrepancies in Table 12 concern supporting claims, not the correct final allowances. Their severity differs; they establish neither an internal mechanism nor a population frequency.

4.5 Completion claims and rendered artifact failures

Two original Claude public-summary responses explicitly described their files as two pages, while PUBLIC-03 contained three pages and PUBLIC-07 four. The additional PUBLIC-06 attempt also produced three pages while claiming two. These are directly inspectable mismatches between the completion message and the delivered artifact. They are not evidence of deliberate deception: the record supports an inaccurate claim, not an inference about intent.

Other files satisfied the page-count check yet remained difficult or impossible to read. GPT PUBLIC-08 overlaps the 80% and 100% figures and neighboring explanations. GPT PUBLIC-01 places a concluding paragraph in dark text on a navy background. Claude PUBLIC-08 draws table headings in the same navy color as their background; the headings remain extractable despite being invisible. Figures 3 and 4 show selected crops from the saved original outputs. Contrast concerns in GPT PUBLIC-09 and PUBLIC-10 are recorded separately, without retrospectively imposing a numeric contrast threshold.

FIRST-PASS	80% 100% +20 pp	12 of 15 retrieval items resolved before
AFTER RETRY		any retry; 13 of 15 resolved after "try a different approach."
RECOVERED		Three first-pass failures became successes without new facts or tools.

Figure 3. GPT PUBLIC-08 original output, page 1. The 80% and 100% figures collide with each other and neighboring text. This crop reproduces the delivered defect; the PDF has two pages and retains extractable text.

Bottom line: AI evaluation should assess not only whether a system can eventually produce a correct answer, but whether it reliably persists, verifies, escalates, finalizes, and respects the boundary between the working conversation and the shareable artifact.			
Retry-probe (Sec. 7)	15-item retrieval; first-pass then one retry prompt	12/15 -> 15/15; gap +20 pp; p = 0.0022 on item-class association	Existence of measurable RBFs; recoverability gap as a separable metric
Intra-vendor (Sec. 8)	10 tasks x 3 Anthropic models (Sonnet 4.6, Opus 4.7, Haiku 4.5)	All 3 models: 100% on QI/PPG/EBC/ACM checks; 0% on RETRY_PROXY rubric	Protocol runs end-to-end; RETRY_PROXY failures traced to rubric vocabulary, not capability

Figure 4. Text extraction can retain visually unreadable content. Top: GPT PUBLIC-01 original output, page 2, dark text on a dark background. Bottom: Claude PUBLIC-08 original output, same-color table headings and background. Crops retain source colors; no contrast enhancement was applied.

Every one of the seven completed additional Claude public-summary attempts had an observed artifact defect. PUBLIC-01, PUBLIC-02, PUBLIC-04, and PUBLIC-10 had table collisions or overflow; PUBLIC-05 had missing mathematical glyphs; PUBLIC-06 exceeded two pages; and PUBLIC-09 had missing glyphs and off-page table text. Six nevertheless met the two-page count. These are selected previously interrupted slots, so this seven-case pattern must not be presented as a general vendor error rate. A terminal answer and a file that opens are insufficient acceptance criteria for a publication task.

Interrupted artifacts remain secondary evidence because the execution cutoff may have prevented repair. They are not counted as completed-response failures.

4.6 Exploratory context attribution and negative findings

Five GPT cost replies (COST-03, COST-04, COST-05, COST-08, and COST-09) described a prior figure as a per-day assertion that should have been monthly. The quoted answer did not contain that per-day line: it gave \$2-4/user in a monthly-plan discussion, without explicitly stating the period on that line. The user challenge introduced the per-day attribution. For example, COST-04 says the "\$2-4 per day" line was wrong. The observable defect is acceptance of an inaccurate description of the quoted history. This annotation was identified after collection began and remains exploratory; it does not establish sycophancy as the cause.

Claude MOBILE-06 supplies a smaller reporting mismatch: its completion message says 14 entries while the artifact contains all 15. This is not a missing-record failure. More generally, the negative findings matter: all final cost allowances and local lookup rows were correct, all completed PDF repairs met the scoped checks, and all mobile cases passed their scoped checks. All five exact mobile duplicate pairs passed in both systems. Their consistency does not validate unrelated historical claims about native phone rendering or live retrieval.

Study 2 supports separate checks of task success, explanation consistency, visual usability and completion claims. It has no exact-input repetition of a completed positive failure, no estimate of recovery after checking those defects, and no independent human validation.

5 Direct API factorial experiment

5.1 Design and collection

The follow-up tests whether the Study 1 configuration asymmetry persists through direct provider APIs and whether repeating the output contract mitigates contamination. It reuses the development bank rather than introducing held-out tasks; it is a precollection-frozen follow-up, not an independently preregistered confirmatory study.

Before any direct-API subject output was collected, two previously separate prospective tests were merged into one contemporaneous collection to avoid introducing a collection-period confound between harness isolation and the checking-by-format-restatement experiment. The completed frozen design used the same 18 tasks, four direct-provider API model configurations, one execution, and 72 shared-anchor task-model blocks. Each block contains a fresh A answer and four independent continuations from the identical visible A transcript, for 360 completed response slots in total.

All four continuations share the instruction "Return a complete final answer to the original task." The checking factor adds "Check the prior response against the original requirements before answering." The format-restatement factor adds "Preserve the originally required output format; add no explanation outside that format." The resulting cells are C = neither factor, E = checking only, R = format restatement only, and ER = both. Branch order and block order are frozen before collection.

The requested and returned model labels were claude-haiku-4-5-20251001, claude-sonnet-4-6, gpt-5.6-luna, and gpt-5.6-terra, with 90 completed responses per model. Calls used the OpenAI Responses and Anthropic Messages APIs, with no researcher-supplied system/developer prompt, no tools, and only the original VIS image attachments. OpenAI reasoning effort was medium; Anthropic thinking, temperature, and sampling seed were omitted. The common output ceiling was 8192 tokens; requests, returned IDs,

latency, and token usage were saved. Sequential collection ran on 11 September 2026 from 22:13 to 22:23 America/Los_Angeles (12 September UTC). Three earlier HTTP 429 credit-balance failures produced no subject answers and were archived before the funded retry. The successful collection completed 360 slots without changing the frozen protocol or rerunning completed wrong answers.

The scorer was frozen before collection. Fixed-line tasks identify the expected payload start and required line count so that material before or after the payload is coded as boundary contamination rather than automatically converted into a deliverable failure. HTML is isolated from document start through the final closing tag, with external text coded separately. Final-deliverable content, final-deliverable format, boundary contamination, and whole-response contract success are reported for every cell.

Harness isolation uses E as the historically comparable verification cell. The frozen descriptive rule classifies the Claude-versus-GPT direct-API E contamination gap as surviving if it retains at least half of the historical 40.28-percentage-point gap in the same direction while OpenAI contamination remains at or below 5%, disappearing if the absolute gap is at most 5 points, and mixed/attenuated otherwise. This label is descriptive, not a population significance test; raw per-model and provider-group counts are primary.

The contemporaneous 2 x 2 analysis reports checking without a reminder (E-C), checking with a reminder (ER-R), format restatement without checking (R-C), format restatement with checking (ER-E), and the additive interaction (ER-R)-(E-C). Boundary contamination is the primary factorial outcome; frozen final-deliverable and whole-response success are co-reported. Each contrast is paired over common A anchors. The historical comparison additionally differs in follow-up wording, repetitions, collection period, and scoring implementation, so it does not identify the causal contribution of a product harness.

The frozen package is preserved with completed outputs in Direct_API_360/ in the v1.40 revision evidence bundle. Its FREEZE_MANIFEST.json SHA-256 remains fa4d4883a5ed676e0734c85dbd85a0cf6c2747c6aef7a61b779a10792dd74c45. All 32 frozen files verified before and after collection. The bundle includes raw responses, exact request bodies, per-slot metadata, attempts, frozen scores, analysis, and a separate integrity audit; credentials are excluded.

5.2 Outcomes under the frozen scorer

Table 14 reports the five conditions separately. A and C each pass 64/72 frozen final-deliverable checks. E and ER each pass 66/72; R passes 65/72. Only E has scored boundary contamination, in 20/72 cases (27.78%). Fifteen of those contaminated E cases pass final-deliverable checks and five fail them, so removing contamination would not by itself repair every failed response. The separate postcollection sensitivity results in Section 7.1 revise three of those five contaminated payload failures to passes.

Table 14. Direct-API outcomes under the frozen scorer. Each condition contains 72 task-model responses; final pass requires both content and format. Whole pass additionally requires no boundary contamination.

Condition	Contaminated	Final pass	Whole pass
A	0/72	64/72	64/72
C	0/72	64/72	64/72
E	20/72	66/72	51/72
R	0/72	65/72	65/72
ER	0/72	66/72	66/72

Table 15. Whole-response boundary contamination. Boundary contamination by direct-API configuration. Denominators are 18 per model and condition. No A responses were scored contaminated.

Model	C	E	R	ER
Haiku 4.5	0/18	7/18	0/18	0/18
Sonnet 4.6	0/18	13/18	0/18	0/18
GPT 5.6 Luna	0/18	0/18	0/18	0/18
GPT 5.6 Terra	0/18	0/18	0/18	0/18

The provider-group E gap is 55.56 percentage points (20/36 versus 0/36), compared with the historical product gap of 40.28 points (29/72 versus 0/72). It exceeds the frozen 20.14-point survival threshold with OpenAI E contamination below the 5% ceiling. The operational result is therefore “survives.” This is a descriptive comparison, not a significance test or an estimate that the APIs caused a larger gap.

Matched historical repetition-1 E to API E pairs number 72. The contamination transitions are 48 absent-to-absent, nine absent-to-present, four present-to-absent, and eleven present-to-present. These pairs use only historical repetition 1, whereas the frozen 40.28-point reference uses both historical repetitions. E contamination is distributed across AUD (4), BOUND (4), FIELD (3), MATH (3), BATCH (2), SRC (2), HTML (1), and LOOK (1); VIS has zero. Family counts are descriptive and do not supply independent replications.

Historical-to-API item-level contamination agreement was 59/72, while the group-level asymmetry retained its direction. The affected items therefore were not stable across executions. Because harness, wording and service period also changed, this observation does not isolate stochastic variation, but it limits per-item claims based on a single execution. The matched branches remain necessary for distinguishing aggregate changes from changes within a task.

5.3 Checking and format restatement

Pooled boundary-contamination contrasts are E-C = +27.78 percentage points, ER-R = 0, R-C = 0, ER-E = -27.78, and interaction = -27.78. In the Anthropic group, E-C is +55.56 points and ER-E is -55.56; both contrasts are zero in the OpenAI group. Per-model E-C effects are +38.89 for Haiku and +72.22 for Sonnet, with equal opposite ER-E contrasts. All other contamination contrasts follow the zero C, R, and ER counts. Twenty matched pairs changed from contaminated E to uncontaminated ER with none reverse (exact two-sided McNemar $p = 1.91 \times 10^{-6}$). The 20 discordant pairs are drawn from 18 tasks crossed with two Anthropic configurations, so they are not fully independent and the nominal p should be read as a descriptive summary of a one-directional pattern rather than a calibrated error rate. D02’s 14 discordant pairs instead correspond to 14 distinct tasks within one model configuration under one execution and do not carry this same task-by-configuration clustering problem.

The E-to-ER whole-response improvement is 15/72 (20.83 percentage points): 51 stable passes, 15 fail-to-pass changes, six stable failures, and zero pass-to-fail changes. Final-deliverable transitions differ: 65 stable passes, five stable failures, one fail-to-pass, and one pass-to-fail change, leaving both aggregates at 66/72. A-to-E final-deliverable transitions are five stable failures, three recoveries, one regression, and 63 stable passes; whole-response transitions are seven stable failures, one recovery, 14 regressions, and 50 stable passes. These are frozen scorer outcomes, not independently adjudicated substantive changes.

6 Held-out D02 factorial experiment

6.1 Design and task bank

Study D02 extends the factorial to 30 new multi-surface probe tasks covering 45 failure surfaces, including HTML, SVG, ICS, CSV, JSON, SQL, Markdown, arithmetic, lookup and boundary contracts. The requested direct-API models were claude-sonnet-4-6 and gpt-5.6-luna. Each model supplied its own A anchor; C, E, R and ER were separate continuations of that model’s visible A transcript. Pairing is within model and task, not across models. The design contains 300 response slots, 150 per model and 30 per model-condition cell, with no recorded transport failures.

6.2 Collection epochs and protocol provenance

The bank was authored after the preceding instrument freeze. Task text, rules and branch instructions were reported frozen before D02 collection. The runner was amended during collection to add Anthropic provider routing and then re-frozen. The retained manifest matches the supplied config, rules, tasks and post-patch runner; its SHA-256 is d295535925fe56344ed161ad53eeb32794b74042210259dee20236ac7d81c373. The reported 4/4 match verifies the surviving freeze, not byte-identical runner execution throughout collection.

GPT collection ran on 12 September 2026 from 16:53:17Z to 16:59:46Z using the pre-patch runner. Claude collection ran from 17:20:31Z to 17:27:25Z using the post-patch runner. The patch script adds provider selection, client dispatch, Anthropic response handling and provider metadata. It does not edit the task text, branch strings or content rules. The archived request messages retain the same task and branch strings across epochs, while each model’s assistant anchor differs. Scoring was run once over all 300 slots. The models were not collected through a byte-identical code path or during the same epoch, so their comparison remains descriptive.

The D02 runner uses OpenAI Chat Completions and Anthropic Messages. Temperature and reasoning were left at provider defaults, unlike the explicitly configured OpenAI reasoning effort in the preceding experiment. The output ceiling remained 8192 tokens. Provider request and response records are retained. The available patch and message records support the stated scope of the amendment; the release does not claim that the surviving post-patch runner alone reproduces the earlier executable.

The exact branch instructions are shown together below. Contamination is any non-whitespace text outside the isolated deliverable; its position is recorded independently of content checking. Final-deliverable pass requires the content and format criteria; whole-response pass additionally requires the absence of contamination.

Table D0. Exact direct-API branch instructions. Each branch continues the same within-model A anchor.

Condition	Instruction
C	Return a complete final answer to the original task.
E	Check the prior response against the original requirements before answering. Return a complete final answer to the original task.
R	Return a complete final answer to the original task. Preserve the originally required output format; add no explanation outside that format.
ER	Check the prior response against the original requirements before answering. Return a complete final answer to the original task. Preserve the originally required output format; add no explanation outside that format.

6.3 Contamination and paired suppression

The contamination effect is conditional on the tested model. Claude Sonnet had 14/30 contaminated E responses and none in A, C, R or ER; GPT Luna had none in any condition. The pooled contrast therefore combines an observed effect with an absent effect on this bank. Ten of the 30 tasks use structured payload types on which the extraction rule can report contamination at either boundary; seven contaminations occurred on those tasks and all seven were prefix, with no suffix observed. The remaining 20 tasks use the fixed-line rule, which takes the final N non-empty lines and returns an empty suffix by construction, so suffix contamination on those 200 slots is not measured. The all-prefix result is therefore reported for the subset where both positions are observable, and is undetermined elsewhere. Manual inspection of all seven fixed-line contaminations confirms that each flagged prefix is genuine leading commentary and that the extracted payload is correct, so no reported label or count changes. The fixed-line position limit was subsequently confirmed by two raters on three constructed probes in Study H (§8).

Table D1. Boundary contamination by model and condition, Study D02

Model	A	C	E	R	ER
claude-sonnet-4-6	0/30	0/30	14/30	0/30	0/30
gpt-5.6-luna	0/30	0/30	0/30	0/30	0/30
Pooled	0/60	0/60	14/60	0/60	0/60

Boundary contamination, whole-response scope: any non-whitespace text outside the isolated deliverable. Conditions branch from each model's own anchor response (A). Ten of the 30 tasks use structured payload types on which the extraction rule can report contamination at either boundary; seven contaminations occurred on those tasks and all seven were prefix, with no suffix observed. The remaining 20 tasks use the fixed-line rule, which takes the final N non-empty lines and returns an empty suffix by construction, so suffix contamination on those 200 slots is not measured. The all-prefix result is therefore reported for the subset where both positions are observable, and is undetermined elsewhere. Manual inspection of all seven fixed-line contaminations confirms that each flagged prefix is genuine leading commentary and that the extracted payload is correct, so no reported label or count changes. Contamination is derived from payload isolation and does not consult the content-checking rules; 12 of the 14 contaminated responses scored content = pass.

Table D2. Condition contrasts, percentage points

Contrast	Pooled	claude-sonnet-4-6	gpt-5.6-luna
E – C	+23.33	+46.67	0.00
R – C	0.00	0.00	0.00
ER – R	0.00	0.00	0.00
ER – E	–23.33	–46.67	0.00
Interaction	–23.33	–46.67	0.00

Contamination-rate differences, whole-response scope. The pooled column averages across a model that shows the effect and one that does not; the per-model columns are the interpretable ones. All fifteen values are locked in 05_NUMBERS_LOCK_v2.json under contrasts_percentage_points; the two-decimal figures are roundings of the locked values.

Table D3. Paired transitions, Study D02

Transition	n_{00}	n_{01}	n_{10}	n_{11}	Exact two-sided p
C → E	46	14	0	0	1.22×10^{-4}
E → ER	46	0	14	0	1.22×10^{-4}
R → ER	60	0	0	0	—
C → R	60	0	0	0	—

Paired within model and task. n_{01} = clean → contaminated, n_{10} = contaminated → clean. Exact McNemar on discordant pairs. Suppression under E → ER is complete and one-directional: 14 of 14, none reverse. The C → E and E → ER tests return the same p because their discordant cells are the same size (14 against 0); they are two views of one set of transitions, not two independent confirmations. Where both discordant cells are zero no test is defined and none is reported.

The exact paired tests condition on the observed discordant pairs. They do not turn the constructed tasks into a random sample of user requests or remove dependence among related task structures. The identical C-to-E and E-to-ER results concern the same contaminated E cases and should not be counted as independent confirmations.

A correct anchor did not preclude contamination: 13 of the 14 contaminated E responses followed correct anchors. This excludes an actual anchor error as a necessary condition for the commentary. It does not reveal what the model believed it had found or isolate the internal cause of its response.

Table D5. Anchor-correctness control

	Contaminated under E	Clean under E
Anchor was correct	13	14
Anchor was incorrect	1	2

claude-sonnet-4-6, 30 tasks. Contamination under checking-only is not explained by the model having found a real error to report: 13 of 14 contaminated responses followed an anchor that was itself correct, a rate of 13 of 27 among correct anchors against 1 of 3 among incorrect ones.

6.4 Payload outcomes and qualified non-replication

Table D4. Dual-scope pass rates, Study D02, after instrument audit

Condition	Final-deliverable /60	Whole-response contract /60
A	57	57
C	58	58
E	57	45
R	58	58
ER	58	58

The same 300 responses under two scoring scopes. Final-deliverable pass counts span 57 to 58 of 60 across all five conditions, a range of one case, with no condition standing out; no test of equivalence was

performed and none is claimed. Whole-response compliance falls to 45 of 60 under E alone. Figures are after the instrument audit of Table D6; as originally scored the corresponding whole-response row reads 53 / 53 / 41 / 53 / 54 and the final-deliverable row 53 / 53 / 53 / 53 / 54.

The earlier aggregate payload-quality penalty did not replicate. Under the earlier audit, E passed 69/72 and ER 67/72; in D02 the corresponding audited counts were 57/60 and 58/60. No aggregate payload penalty was observed, and one of 30 tasks showed the earlier direction at the item level, which is consistent with but far too small to support the earlier effect. NUM-01 is that case: checking alone fully repaired its arithmetic, whereas checking with the reminder repaired it only partially. An aggregate comparison can conceal that item-level difference.

Table D6b. Content repair under the two factors

Task	A	C	E	R	ER
D02-NUM-01	3 errors	3 errors	all correct	3 errors	1 of 3 fixed
D02-NUM-04	wrong	wrong	wrong	wrong	wrong

claude-sonnet-4-6, the two tasks with persistent arithmetic error. The checking instruction repaired NUM-01 completely under E and partially under ER, and did not repair NUM-04 under any condition. The E response that carried the corrected figures was itself boundary-contaminated. With two such tasks this supports no general claim about whether checking improves correctness; it is reported because it bears on the payload-quality question and because the opposite claim would otherwise be inferred from Table D4.

For Claude NUM-01, A, C and R retained KX-1 = 69, KX-3 = 62 and KX-F = 78 against the required 77, 70 and 75. E returned the fully corrected sequence 77, 65, 70, 75 with boundary contamination. ER corrected KX-1 to 77 but retained the other errors and was clean. NUM-04 remained wrong under all conditions, returning 415 and 414.9254 against 416 and 415.2985. The checking instruction repaired content in one of the two persistent-arithmetic tasks and not in the other; repair was complete under E and partial under ER. With two such tasks this supports no general claim in either direction.

7 Instrument audits and sensitivity analyses

The audits below distinguish changes to substantive acceptance labels from limits of boundary extraction and rendered-artifact measurement. Frozen results remain separate from postcollection judgments; the subsequent human adjudication addresses the boundary measure.

7.1 Direct API extractor failures and sensitivity scores

An AI-assisted postcollection audit reviewed all 360 responses against the 18 original prompts and two chart images. Within each family, exact responses were grouped into 71 distinct texts, with every slot linked to its task and raw-text hash. The audit checked paired variants against their requirements, examined passes and failures, recomputed payments and budgets, and inspected all distinct HTML sources and both charts. It did not independently validate rendered artifacts. The assistant also revised this manuscript and knew the frozen scores and conditions. This is unblinded sensitivity analysis, not human adjudication; identical texts are not independent replications.

The audit identifies the explicitly presented final answer after checking prose rather than selecting the first repeated label. It accepts an unambiguous equivalent date where the prompt specifies the date but not its spelling. Content and format are recorded separately: a notice that preserves every fact but violates the

prescribed line order fails format, not factual content. These postcollection judgments are documented for all cases and leave frozen results intact. They are not a newly validated general-purpose scorer.

Seven frozen final-deliverable failures become passes: the five Sonnet AUD-scenario answers using “October 12, 2026,” Sonnet BATCH-scenario E whose initial calculations falsely trigger the AL-1 anchor, and Sonnet BOUND-comparison E whose checklist falsely triggers the notice anchor. Four whole-response labels change, all uncontaminated AUD answers in A, C, R and ER. The other three corrected payloads still have prohibited commentary. Across component labels, 21 content labels and eight format labels differ, partly because the frozen BOUND checker encodes line arrangement as content. Seven overall labels change; component differences must not be added as distinct corrected responses.

All seven final-deliverable corrections went from fail to pass, with none in the reverse direction. This directional split is a concrete limitation of an audit conducted with knowledge that the extractor might be too strict. It is compatible with genuine over-rejection, but the audit cannot exclude preferential detection of over-strict labels. A blinded audit need not necessarily find changes in both directions; the present asymmetry is a reason for independent checking, not proof that these corrections are wrong.

Table 16. Final-deliverable and whole-response scope. Direct API postcollection AI-assisted sensitivity scores alongside unchanged frozen scores. Each condition has 72 responses. These are unblinded judgments, not independent human labels.

Condition	Final frozen	Final audit	Whole frozen	Whole audit
A	64	65	64	65
C	64	65	64	65
E	66	69	51	51
R	65	66	65	66
ER	66	67	66	67

Boundary contamination agrees in all 360 cases: E has 20/72 and every other condition has zero. All 20 contaminated responses have checking or explanatory prefixes; the frozen prefix/suffix position labels can be artifacts of erroneous extraction. Under audit scoring, adding the reminder to E yields 16 whole-response fail-to-pass changes, 51 stable passes and five stable failures, with no whole-response regressions. Final-deliverable transitions instead have 67 stable passes, three stable failures and two pass-to-fail changes, Sonnet BATCH-scenario (a payment error) and Sonnet BOUND-scenario (a line-order violation). Thus E-to-ER final acceptance decreases from 69/72 to 67/72 while whole-response acceptance rises from 51/72 to 67/72. A-to-E final transitions contain four recoveries and no regressions; whole-response transitions contain one recovery and 15 regressions. These paired counts are descriptive and depend on the stated audit judgments.

7.2 Payload quality and contract compliance

Under the unblinded audit labels, final-deliverable acceptance ranked as follows:

- E: 69/72
 - ER: 67/72
 - R: 66/72
 - A and C: 65/72

Adding the reminder produced two payload pass-to-fail changes and no fail-to-pass changes, while whole-response passes rose from 51/72 to 67/72. This fixed-bank tension supports distinguishing machine consumption of the whole response from human use of an isolated answer. It does not establish a general payload cost, as the D02 aggregate comparison in Section 6.4 does not reproduce it.

The separate unscored review packet supports a future human check of every response, with metadata masked and original prompts and chart images provided. Two reviewers should label independently before adjudication; the audit judgments and mapping should be withheld until their initial ratings are locked. Response wording can reveal the checking condition, so metadata masking cannot guarantee full blinding. The present analysis establishes neither inter-rater reliability nor an independently validated scoring standard. The fixed development bank, one execution per model-task cell, provider-model confounding, service defaults, collection dates and historical prompt differences continue to limit generalization and causal interpretation.

7.3 D02 content scoring and defect reconciliation

The D02 audit re-examined the automated failures against the saved responses and preserved the original scores. It identified date-string literalism, a hardcoded SVG coordinate assumption and entity matching against raw HTML. These are acceptance-instrument defects: the checked representation can reject a response for reasons not required by the task. Defect instances and slots corrected to pass are different units, because an affected slot can retain an independent failure.

Table D6. Instrument audit of content scoring, render checking and boundary detection limits

Defect in the checker	What the model actually produced	Instances	claude / gpt
Date literalism	"Wednesday, 12 August 2027" — correct date, prose form	19	15 / 4
SVG coordinate system hardcoded to 0–200	width="200" viewBox="0 0 3 1", three unit-width rects: an exact tiling	4	0 / 4
Entity resolution on raw source	№7, which renders as №7	1	1 / 0
Total content-defect instances		24	16 / 8
Render checker selected elements by descendant text, then tested the container's own inherited colour against its own background.	a <header> with its own background whose visible text is a differently coloured child <h1>	1	1 / 0
Fixed-line extraction returns an empty suffix by construction	a trailing sentence after a three-line answer is absorbed into the payload window and labelled prefix	No observed case in the 300 collected responses; confirmed on 3 constructed probes by 2 raters (Study H)	Detection limit: 200 of 300 slots

Table D6 (continued). Reconciliation to slots

Reconciliation to slots	Count	claude / gpt
Failing slots before audit	34	26 / 8
Slots flipped to pass by the correction	22	14 / 8
Slots carrying a defect that still fail for an independent reason	2	2 / 0
Genuine model errors remaining	12	12 / 0

Every failure the automated scorer reported was re-examined against the saved response text. Defect instances (24) exceed flipped slots (22) because two slots carried both a checker defect and an independent genuine failure: D02-AUD-01 under E, which also leaked excluded tokens, and D02-AUD-02 under A, which also broke a three-line format contract. Both still fail after correction. Twenty-two of thirty-four failures were therefore defects in the measuring instrument rather than in the responses, across both models. The instrument audit spans five instrument defect classes; the render-check instance and fixed-line detection limit are separate from the 34 content-scoring slot failures. The fixed-line row describes a detection limit, not an observed miscount. The content reconciliation remains 24 defect instances, 22 flips, two retained independent failures and 12 genuine errors. Correcting them moves final-deliverable pass rates from 124/150 to 138/150 and from 142/150 to 150/150 respectively — the correction narrows rather than widens the between-model gap.

The audited final-deliverable totals remain sensitive to a declared SVG tolerance. A GPT response used `width="33.33333333333333%"` repeatedly, leaving a 2×10^{-13} -unit gap. The audit accepts it at a 0.1% tolerance; strict scoring would instead give GPT 149/150. The main audited table retains the supplied tolerance-based labels, and this stricter sensitivity is not substituted into the frozen scores.

The content audit does not independently validate payload extraction: the contamination flag was computed separately and was unchanged by these corrections. Twelve of the 14 contaminated responses had passing content. Conversely, repeatable extraction does not validate the content checker. The earlier fixed-line and HTML mutation checks establish their documented boundary behavior only; they are not a general validation of the D02 acceptance rules.

Independent validation of content checking and rendered-artifact assessment remains unresolved. The D02 audit was conducted with access to saved outputs and machine outcomes, and the current revision does not present it as blinded human adjudication. Study H reports human agreement with the extraction-based boundary flag, while substantive scoring and rendered-artifact assessment require their own evidence.

7.4 Render checker and corrected measurements

Table D7. Corrected contrast measurements for D02-HTML-01

Condition	claude-sonnet-4-6	gpt-5.6-luna	Threshold
A	5.65 : 1	13.10 : 1	4.5 : 1
C	5.65 : 1	13.10 : 1	4.5 : 1
E	5.65 : 1	13.10 : 1	4.5 : 1
R	5.65 : 1	13.10 : 1	4.5 : 1
ER	7.86 : 1	13.10 : 1	4.5 : 1

Twenty HTML slots were rendered. Ten carried a contrast requirement, all from one task; the other ten were checked for horizontal overflow only and none overflowed. The minimum across the ten contrast-checked slots after correction is 5.65 : 1.

The corrected minimum contrast for claude-sonnet-4-6 D02-HTML-01 ER is 7.86 : 1 against the 4.5 : 1 threshold. The worst element is a navigation anchor at #1a1a1a on #e8a020. The recorded 1.49 : 1 came from a <header> painting no direct text, tested using the inherited body colour #1a1a1a against its own background #1a3a5c; the text actually painted inside it is an <h1> at #ffffff. The corrected selector includes elements having at least one direct non-empty text child node. A, C, E and R on that task measure 5.65 : 1 under both selectors, so this defect is triggered only by the header-with-background pattern used by ER. The author re-measured all ten D02-HTML-01 slots, covering both models and all five conditions, at 1280 × 800 in the same browser engine with the original and corrected selectors run side by side. Re-measuring all ten contrast-checked slots with the corrected selector leaves none below the threshold, so D02 recorded no true contrast failure. The corrected selector has been validated against this task only; contrast checking on other page structures, and every other rendered-artifact property, remain untested.

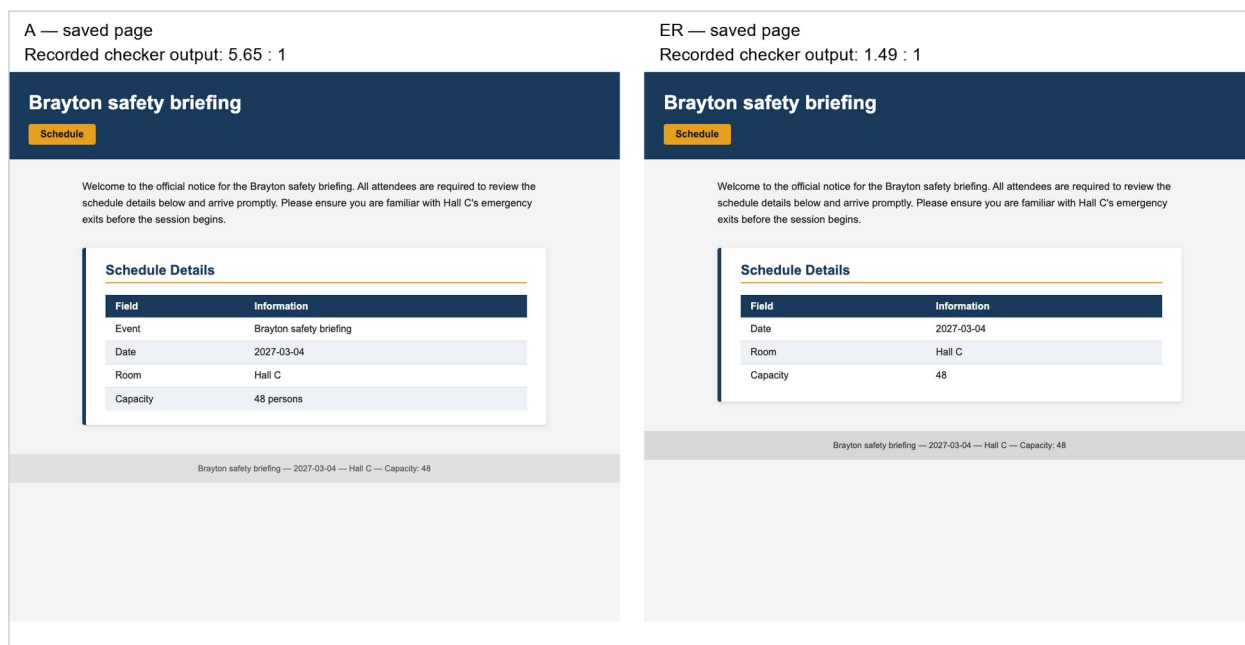


Figure 5. Claude D02-HTML-01, saved A and ER pages rendered at the same viewport. Source colors and content are unchanged. Image labels preserve the original checker outputs for provenance; Table D7 reports the corrected measurements. The apparent ER failure was a render-checker defect, not a page defect.

8 Human adjudication of the boundary measure: Study H

8.1 Question and rater disclosure

Study H tests whether the automated boundary-contamination flag matches human judgement. Two independent raters each adjudicated the same 47 masked cases, without access to the sealed key, the model names, the condition codes or the slot identifiers. Each case presented the output-format requirement and the complete response, and asked whether there was any non-whitespace text outside the required deliverable and, if so, where. Raters were told explicitly that correctness was not being judged.

Rater disclosure. Both raters were independent of the author and the work. Neither had prior knowledge of the paper before rating.

8.2 Packet construction and preregistration

The packet shuffled two strata together without displaying their labels. Stratum A contains 30 real D02 responses: a census of all 14 machine-positive cases and 16 machine-negative controls across both models and 16 tasks. Stratum B contains 17 minimal perturbations of real payloads. These probe trailing commentary, sign-offs, code fences, comments after closing tags or inside documents, trailing whitespace, and commentary without a deliverable. The frozen isolate() generated their machine labels. Case order used seed 20260913, and the key hash was committed before rating. Raters knew some responses were edited to probe boundaries, but not which.

Preregistration. For stratum A the primary measure is Cohen’s kappa between the human yes/no judgement and the automated flag, with a corroboration criterion of kappa at least 0.80 and no more than two disagreements. For stratum B the preregistered analysis is descriptive: exact agreement on the yes/no judgement and on the reported position, with no numeric threshold. Unsure responses are reported as a count and excluded from kappa. The analyzer output is released in full, with every disagreement named. The revised packet supersedes the earlier unrated packet; the original materials are retained as provenance.

8.3 Agreement on real responses and constructed probes

Table H1. Study H agreement

Stratum	Rater	Rated	Q1 agreement	Kappa	Position agreement
A (real)	1	29	29/29	1.000	14/14
A (real)	2	29	29/29	1.000	14/14
B (probes)	1	17	16/17	0.866	8/12
B (probes)	2	15	15/15	1.000	6/9

Rater 1 recorded one unsure response (H04); rater 2 recorded three (H02, H16, H40). These are excluded from kappa and listed in Appendix E. Kappa on stratum A has a reported 95% interval of [1.000, 1.000]; rater 1’s stratum-B kappa has an interval of [0.612, 1.000], and rater 2’s is [1.000, 1.000]. These are the analyzer’s approximate intervals. Its variance estimate is zero at perfect observed agreement, so a collapsed interval does not establish absence of uncertainty beyond these cases. Position agreement uses definite human-positive responses as its denominator, including a human-positive/machine-negative disagreement; the analyzer’s label “shared positives” is therefore inaccurate.

Table H2. Agreement between the two raters

Stratum	Exact agreement	Kappa
A (real)	28/30	1.000
B (probes)	14/17	0.857

Exact agreement in Table H2 includes unsure responses as a response category, whereas kappa excludes cases where either rater was unsure. Both stratum-A differences are one rater’s unsure against the other’s

definite answer (H02, H04); neither is a yes-versus-no conflict. The three stratum-B differences are H11 (yes versus no) and H16 and H40 (yes versus unsure).

On the real cases, both raters agreed with the automated flag on every case they rated, including the reported position of the contaminating text. This corroborates the labels on the real cases adjudicated. It is a ceiling result on an unambiguous set: the collected D02 responses contain no code fences, no trailing commentary and no borderline cases in the supplied audit, so stratum A cannot distinguish a robust rule from a rule that happens to fit easy inputs. It is reported as corroboration of the reported labels, not as evidence of general robustness.

8.4 Replicated position disagreements and the defective probe

Both raters disagreed with the rule on the same three position judgments, all fixed-line probes. They placed the trailing sentence (H21) and sign-off (H39) after the deliverable; the rule labels both before. They placed the fence around H25 on both sides; the rule reports before only. These reproduce the detection limit in §6.3: final-N-line extraction absorbs trailing text and returns an empty suffix. Previously established from the code, the limit is now corroborated by two raters who, according to the study report, saw neither the code nor the probes' specific purposes.

The yes-or-no decision is unaffected on those three probes. Across both raters and both strata, the only yes/no disagreement with the rule is H11, and H11 is a defective probe: it removes a required line, so it tests completeness rather than the presence of extra text, which is not what the rater question asks. The two raters split on it. It is retained in the reported calculations but excluded from the probe-level interpretation; it is evidence about the probe rather than about the rule.

8.5 Validation limits and preregistered outcome

Study H addresses the boundary flag and position labels on the adjudicated cases, not every extraction behavior or the content and render checkers. Human-human kappa of 0.857 on probes shows disagreement on hard boundaries and limits a human-referenced interpretation. Two notes judged correctness instead of extra text; one misread a single calendar event block as two events. This shows drift despite the instructions. After rating, one analyzer line added “Before and After” as a synonym for “both”. This changed rater 2's probe position agreement from 5 of 9 to 6 of 9 and affected no other reported figure.

Preregistered outcome. The stratum-A criterion was met for both raters, with kappa 1.000 and zero disagreements among definite judgments. Stratum B was preregistered as descriptive and is reported as such.

9 Related work

9.1 LLM benchmarks and evaluation

HELM (Liang et al., 2022), BIG-Bench (Srivastava et al., 2023), and MMLU (Hendrycks et al., 2021) cover reasoning, knowledge, and language tasks. Their primary targets are terminal-output correctness, calibration, or preference under fixed prompts. Headline scores alone do not report paired transitions, although extensions can evaluate retries and interaction. This paper separates first-pass from post-retry performance and artifact readiness from text correctness.

GenRec illustrates why endpoints must remain separate: it reports approximately 1.6% offline MRR lift, 0.115% short-term engagement lift and 0.006% long-term core-metric lift (Li et al., 2026, Section 5.1).

These are different metrics. Their ratio measures neither lost performance nor recoverable failure; the example motivates separating deployed outcomes from offline scores.

9.2 RLHF and preference learning

Reinforcement learning from human feedback uses preferences over candidate responses to guide training (Christiano et al., 2017; Bai et al., 2022). A preference label identifies the preferred answer but does not itself explain why, or whether a dispreferred answer would improve after a retry without new information. Aggregate preference therefore need not capture failures whose cost appears in additional supervision turns.

9.3 Tool-use and agentic benchmarks

SWE-bench (Jimenez et al., 2024) and WebArena (Zhou et al., 2024) measure end-to-end task completion with tools in software engineering and web navigation. They do not always separate intermediate decisions to retry a search, change a query, verify a claim, request visual clarification, or declare an artifact complete. Paired transitions and artifact checks expose failures at those decisions.

WebGPT (Nakano et al., 2021), Gorilla (Patil et al., 2023), ToolLLM (Qin et al., 2023), and Voyager (Wang et al., 2023) provide retrieval, API-use, and extended-interaction settings. They motivate recording tool affordances and acquired evidence, but do not validate this paper's observation bank.

9.4 Truthfulness, sycophancy, and deception

TruthfulQA tests false-but-popular claims (Lin et al., 2022); sycophancy work examines agreement with user preferences that diverge from correct answers (Sharma et al., 2023). Park et al. (2024) survey deception and adjacent confidently-wrong behavior. The present observation bank also covers refusals, truncation, misreading, and incomplete artifacts. Such failures can impose workflow costs without taking the form of a false assertion.

9.5 Time-sensitive QA and knowledge staleness

Time-sensitive QA documents how static knowledge degrades as the world changes (Chen et al., 2021; Liska et al., 2022). Procedural guidance can likewise cease to match a third-party interface. Evaluation requires dated sources and inspection of the actual interface: a mismatch alone cannot distinguish stale training exposure, unsupported guessing, and product change.

9.6 Long-horizon agent reliability and human-AI workflow reliability

MINT and tau-bench evaluate settings where intermediate decisions, feedback, and repeated execution affect success (Wang et al., 2024; Yao et al., 2024). Users may also spend time detecting premature stopping, reissuing requests, and auditing artifacts. This paper does not quantify that burden or establish a universal decline with task length. Both questions require measured time, traces, and a defined sampling frame.

Search, verification, escalation, finalization, and audit are candidate first-pass decisions within a workflow. Their sequence and dependence matter; multiplying marginal success rates requires additional assumptions. Separate first-pass and retry rates, alongside artifact and boundary checks, make these outcomes measurable within declared, frozen configurations.

9.7 Case-study methodology

Following Yin's (2018) theory-building case-study logic, naturalistic workflow cases identify constructs for controlled replication. They do not replace experiments. The historical retry pilot in Appendix B

illustrates this measurement step; the direct-API factorial in Section 5 extends it with a protocol frozen before collection.

9.8 Self-correction and repeated-trial reliability

Self-Refine iteratively improves outputs through model feedback (Madaan et al., 2023); Reflexion uses verbal feedback in agent learning (Shinn et al., 2023). Improvement after an initial response is therefore established. Huang et al. (2024) report limitations and degradation in intrinsic reasoning self-correction, while Kamoi et al. (2024) distinguish feedback sources and conditions supporting correction. Study 1 uses fixed, non-answer-bearing follow-ups without correctness labels and measures both benefit and harm.

MINT tests multi-turn problem solving with tools and language feedback (Wang et al., 2024); τ -bench tests tool-agent-user interaction and repeated-trial reliability (Yao et al., 2024). Certainty Robustness distinguishes justified corrections from unjustified changes under challenge prompts (Saadat and Nemzer, 2026). Neither two-turn evaluation nor the correction/regression distinction is novel here. The proposed combination adds heterogeneous workflow acceptance tests, a shared-anchor fresh-repeat comparison, and intervention costs.

Chain-of-thought prompting (Wei et al., 2022), ReAct (Yao et al., 2023a), and Tree of Thoughts (Yao et al., 2023b) alter reasoning or action structure. Their effects do not estimate the fixed verification prompt used here. Comparisons must specify the intervention, feedback, and resources.

Cemri et al. (2025) derive a multi-agent failure taxonomy with expert annotation and agreement measurement, a standard this single-researcher observation bank does not yet meet. ToolMisuseBench tests tool misuse and recovery in deterministic environments with explicit budgets (Sigdel and Baral, 2026). Expected Recovery Regret formalizes recovery-policy performance under execution noise (Vuddanti and Chittiprolu, 2026). These operational faults and policies differ from a standardized conversational follow-up to a complete but unsuccessful deliverable.

Table 13. Related work and the scope of the present workflow evaluation.

Work	Shared concern	Boundary of this proposal
Self-Refine; Reflexion	Improvement after feedback	Fixed follow-ups; no answer or correctness feedback supplied
Huang et al.; Kamoi et al.	Limits of intrinsic correction	Explicit fresh-repeat control and harm accounting
MINT; τ -bench	Interactive success and reliability	Per-anchor directed transitions plus artifact acceptance
Certainty Robustness	Corrected and unjustified changes	Workflow tasks and verifier-grounded operational cost
MAST; tool-recovery work	Failure classes and recovery	Exploratory hierarchy; system faults separated from model outcomes

We claim neither priority for the acronym RBF nor an exhaustive literature search. The contribution is a measurement specification applied to workflow acceptance problems, with study-specific validation limits.

IFEval operationalizes verifiable instructions (Zhou et al., 2023). It supports testing explicit constraints but does not validate this study's lexical or semantic checks. Artifact acceptance and content correctness remain separate outcomes.

10 Discussion and limitations

10.1 What the completed studies establish

Study 1 separates acceptance scopes: verification produced 29 whole-response contract violations, while all 29 final payloads pass the deterministic rescore. It also repaired two repeated arithmetic failures. Within this bank, improved payloads coexisted with 29 responses that were less clean for direct consumption.

Study 2 shows that correct allowances, successful lookups, and required page counts can coexist with inconsistent explanations, inaccurate completion claims, text collisions, missing glyphs, or invisible text. These findings support separate acceptance dimensions rather than a single terminal score.

Recoverable behavioral failure remains useful for directed transitions, but the two observed recoveries repeat one arithmetic task. The stronger evidence concerns output-contract scope and artifact readiness. The sixteen-label bank remains a hypothesis generator.

10.2 Threats to interpretation

The development banks target particular patterns rather than ordinary user requests. Study 1 has nine paired templates and two repetitions; Study 2 has related price, audience, and layout variants, including five exact duplicate pairs. Large counts and equal family allocation do not ensure independence or representativeness. Neither development study claims preregistration or independence-based population intervals. Nine families create dependence, but neither nine nor 576 is automatically an effective sample size. Family counts describe concentration. Variance or design-effect estimates require an explicit sampling and dependence model with adequate replication.

The author formed the hypothesis and developed both the earlier and D02 banks with AI assistance. D02 tasks were written after the preceding instrument freeze and reportedly frozen before collection. They are held out from the earlier bank, not from the hypothesis or investigator. Section 6.2 discloses the runner amendment and surviving freeze evidence.

Claude Code and Codex CLI supplied different defaults and residual context, confounding the original product contrast with harness. The direct-API experiment reproduces the direction of the checking-only asymmetry without those harnesses (Section 5). This weakens a specifically Claude-Code-versus-Codex explanation but does not separate model weights, provider behavior, reasoning defaults, aliases, or collection period. Checking wording also changed, preventing a causal estimate of harness removal.

Study 1's codebook followed collection; its dual-scope rescore followed external criticism. Deterministic extraction limits further discretion but does not make that analysis preregistered or independently validated. Study 2 remains unblinded, without independent human labels. The direct-API scorer was frozen before collection but designed with knowledge of Study 1. Freezing did not prevent label repetition from misdirecting extraction or the AUD checker from rejecting an equivalent date spelling. Section 7 reports these instrument failures while retaining frozen scores.

Study 1's original audit used the author and one novice contact. It encountered one of the 29 whole-response regressions and missed all six arithmetic-error outputs. The later 36-output follow-up broadened coverage but had only 18/36 format agreement with frozen screening. A masked 60-case packet covers all

29 scope-sensitive regressions, both recoveries, and 29 passing controls; new Study 1 human scores remain pending. D02's primary measure is directional: leading contamination is detectable across the bank, trailing contamination only on the structured-payload third. Fixed-line suffix position is unmeasured by construction. This affects payload isolation and the primary outcome, not merely content scoring. Study H corroborates real-case boundary labels and confirms the position limit on constructed probes (§8); it does not independently validate content or rendered-artifact checks.

Some historical raw results and request traces are unavailable, and several lexical tests did not measure task success (Appendix B). Later studies and related work cannot validate missing execution evidence. PUBLIC tests transformation and delivery of the author's v1.23 manuscript, not its scientific claims or generalization to other documents. Faithfully repeating a source flaw differs from introducing an unsupported claim. Session memos likewise provide cases, not external replication.

10.3 Implications for evaluation and practice

Define the object being accepted. Preserve the task, complete response, final payload or artifact, completion claim, and bounded continuation. Score content, whole-response compliance, boundary contamination, artifact readability, and source support separately where the workflow requires those distinctions.

The reminder suppressed observed contamination in both factorial experiments, supporting its use when the complete response must meet an output contract. Payload consequences differ: the earlier audit gave E 69/72 versus ER 67/72, while D02 gave 57/60 versus 58/60, with NUM-01 retaining the earlier direction. Neither a general correctness benefit nor a general penalty follows. Payload extraction needs a validated extractor; whole-response consumption also needs a contract check. Study 2 supports inspecting rendered artifacts.

10.4 Remaining Study 1 and direct API adjudication

The original AI screen, deterministic scorer, and unblinded AI audit are successive checks within one investigator-led process. Their accumulation does not constitute independent validation. Study H addresses the narrower extraction question (§8); content and rendered-artifact checks still require independent evidence.

The public archive includes a masked 60-case packet: all 29 original whole-response A-to-E regressions, both recoveries, and 29 passing E controls selected with a fixed seed. All 60 cases are E responses; the two A-side \$63 anchors are absent, so only the E side of each transition is adjudicated. Controls are matched by task family, but 19 of the 29 come from GPT configurations because comparable uncontaminated Claude passes were scarce. Response style may reveal the stratum. Raters score whole-response compliance, final-deliverable content, final-deliverable format, and boundary contamination separately. The private answer key should be released only after scoring; its SHA-256 commitment is public. Raters should receive the packet alone: matching its text against public Study 1 answers could defeat metadata masking.

At least two raters without prior exposure to the results should score the packet independently before adjudication. Report agreement by dimension and retain disagreements. Until then, the dual-scope rescore remains sensitivity analysis rather than independent ground truth. Study 2 also needs independent rendered-artifact review.

10.5 Remaining validation scope

For incident-derived work, retain original attachments, output surfaces, client or search conditions, and relevant preceding turns where permission allows. Document substitutions. Calibrate difficult tasks separately from held-out evaluation tasks. Prevalence claims require a second bank sampled from a defined ordinary-use population; the current constructed banks support mechanism discovery and acceptance design, not population error rates.

Future confirmatory work should freeze tasks, scoring scopes, model/harness identifiers, exclusions, analysis plans, and dependence treatment before collection. Precision calculations cannot repair biased labels or unrepresentative tasks. Independent scoring and faithful execution context remain priorities.

11 Conclusion

The observed contamination effect is model-conditional. The earlier product and direct-API studies retained a configuration asymmetry; on D02, checking-only contamination occurred in 14/30 Claude Sonnet responses and 0/30 GPT Luna responses. Adding the reminder suppressed 14 of 14 paired contaminations, with none reverse. The within-model intervention comparison does not identify a causal model or vendor difference across collection epochs.

The earlier aggregate payload penalty did not replicate: audited E-to-ER final-deliverable passes were 69/72 versus 67/72 in the preceding experiment and 57/60 versus 58/60 in D02. NUM-01 retained the earlier direction at the item level, with complete arithmetic repair under E and partial repair under ER. The instrument audit further shows why content, extraction and whole-response labels must remain distinguishable: checker defects changed acceptance decisions while contamination labels remained unchanged. Supporting Study 2 artifacts retain the need for rendered inspection. Study H corroborates the real-case boundary labels and confirms the fixed-line position limit on constructed probes. Independent validation of content checking and rendered-artifact assessment remains outstanding.

Data, code, and AI assistance

The v1.45 revision package preserves the v1.40 baseline and adds the D02 instrument, raw response export, original and audited scores, render-check record, and completed Study H packet, both scoring sheets, sealed machine key, committed hash manifest and analyzer. The underlying v1.40 revision evidence bundle and v1.38 research archive remain the sources for the earlier studies. The current release includes the corrected number lock and supplied tables, a manuscript change record and a numbers audit. D02 request records and the provider patch are retained as collection provenance. The Study H machine key is released after rating; other private rater keys and credentials are excluded. Original responses, frozen scores and instrument files are not overwritten.

Study 1 still includes 18 prompts, two visual inputs, 576 responses, branch metadata, frozen screening labels, and de-identified historical human ratings. Study 2 includes the 100-slot ledger, original and additional attempts, generated artifacts, requirement checks, fidelity notes, and collection amendments. File hashes and reproduction scripts support accounting checks without claiming that reproducibility of a label makes the label true.

Generative AI assisted with task construction, collection, screening, sensitivity analysis, artifact inspection, protocol implementation, and writing. The v1.40 API sensitivity judgments were made by the same assistant revising the manuscript, with prior access to models, conditions and frozen scores. They are an unblinded AI-assisted audit, not independent human validation or a second independent rater. The

author remains responsible for the claims and release. The API collection is complete; the separate human review packet for the preceding direct-API experiment is prepared but unscored. Study 1 independent adjudication and Study 2 independent rendered-artifact review also remain outstanding. The D02 content audit and the present revision remain part of the author-coordinated, AI-assisted process. Study H is complete and reports two raters’ judgments on the boundary measure; both were independent and had no prior knowledge of the paper, as disclosed in §8.1.

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Appendix A: Observation bank and operational boundaries

The sixteen labels below organize earlier author-reported patterns. They are not sixteen experimentally validated or mutually exclusive model defects. The unit is one task attempt and bounded continuation. Code the failed requirement first; assign a candidate workflow label only when its operational boundary is supported. System/client conditions and evaluation omissions remain separate layers. Possible internal causes and mitigation benefits require further evidence.

Table 17. Provisional observation labels, organizing layers, and possible user costs; causal attribution requires evidence.

Code	Name	Candidate locus / layer	User-visible cost
PTS	Premature Termination of Search	Model / retrieval policy	False unresolved answers
BDTF	Breadth-Depth Tradeoff Failure	Model policy under batch load	Shallow per-item analysis
SMI	Search Methodology Inconsistency	Retrieval strategy	Inconsistent escalation
MRT	Mid-Response Truncation	Generation / transport; unresolved cause	Lost progress
MSI	Mobile Streaming Interruption	Client / transport	Dropped responses
FPG	Feedback Persistence Gap	Product / training pipeline	Corrections not preserved
RFI	Retry-Failure Invisibility	Evaluation pipeline	Missing transition evidence
STK	Stale Third-Party Tool/UI Knowledge	Model knowledge / tool guidance	Confident wrong UI paths
QI	Quantitative Inconsistency	Reasoning / self-audit	Contradictory numbers
PPG	Phantom Placeholder Persistence	Artifact / code generation	Placeholders as commands
UCT	Usage-Cap Termination	System / rate limit	Degraded resume state
VMC	Visual Misreading Cascade	Multimodal interpretation	Cascading misreadings
AFF	Artifact Finalization Failure	Artifact generation	File exists but unusable
SVD	Source Verification Drift	Citation / retrieval practice	Weakly supported claims
EBC	Editorial Boundary Contamination	Artifact boundary management	Process notes leak in
ACM	Audience / Context Miscalibration	Communication / register	Off-register content

Adjacent labels distinguish the unit and evidence needed: PTS concerns resolution after retrieval escalation, whereas SMI compares strategy across comparable subtasks; MRT has an unresolved interruption source, MSI requires client evidence, and UCT identifies a quota event. AFF concerns artifact readiness, PPG a missing concrete value, and EBC prohibited editorial content. These distinctions may be merged or revised after independent annotation. They are not justified by assuming an unobserved causal layer.

Premature Termination of Search (PTS)

Definition. An item is declared unresolved, then verified as resolved after an eligible retry using an alternative retrieval strategy. Search traces can establish the observed query change; they do not prove that the initial attempt had exhausted or deliberately withheld available strategies.

Observed pattern. In the historical retrieval pilot, generic queries left three opaque-identifier URLs unresolved; a standardized retry used site-targeted queries and resolved them (Appendix B). Additional search effort, evidence, and inference computation remain alternative explanations.

Operational boundary. Unresolved item followed by verified resolution after a fixed retry; inspect search trace. A possible untried strategy alone does not establish recovery.

Evaluation and proposed mitigation. Record queries, retrieved evidence, and verified outcomes before and after a bounded alternative strategy. Test escalation policies prospectively.

Breadth-Depth Tradeoff Failure (BDTF)

Definition. Under increased batch load, per-item verification may weaken while apparent coverage remains complete; an internal optimization policy is a hypothesis, not directly observed. The result is a superficially complete response with weak per-item grounding: every item appears addressed, but per-item depth degrades as the batch size grows.

Observed pattern. Multi-item evaluations give similarly brief, confident dispositions to items of unequal difficulty, sometimes without required secondary lookup. Uniform output length can conceal unequal verification depth.

Operational boundary. Per-item verification weakens under batch load. Test balanced batch-size conditions; do not infer causation from long output alone.

Evaluation and proposed mitigation. Use balanced batch sizes on the same tasks and record per-item verification and correctness. Report shallow or unverified items explicitly. Study 1 did not establish the hypothesized batch-induced error.

Search Methodology Inconsistency (SMI)

Definition. The model uses stronger retrieval tactics for some items and weaker tactics for similar items within the same session, with no explicit reason for the difference. Query construction varies between generic keyword search, site-targeted search, and identifier-anchored search without a stated escalation rule.

Observed pattern. Comparable retrieval subtasks receive different escalation strategies: a tactic succeeds on one item but is not attempted on another. PTS and SMI may overlap in such a sequence.

Operational boundary. Different escalation across comparable subtasks; requires query logs and comparability. Strategy variation can be appropriate.

Evaluation and proposed mitigation. Compare query construction and escalation across genuinely comparable subtasks. A documented escalation policy is a proposal, not evidence that uniform tactics are always optimal.

Mid-Response Truncation (MRT)

Definition. The model begins generating a response, makes partial progress, and the stream terminates before completion, often after partial tool use or partial drafting. The termination is signaled by a generic error rather than a graceful wind-down.

Observed pattern. A long, heterogeneous response terminates with a generic recovery message. Partial work is not preserved or replayable, and a fresh prompt lacks the intermediate state, adding recovery work as task complexity grows.

Operational boundary. Incomplete generation/stream; distinguish token-limit stop, transport fault, and voluntary stop through telemetry. Not automatically model-core behavior.

Evaluation and proposed mitigation. Retain stop reasons, token/tool use, transport logs, and checkpoints. Distinguish resource enforcement and transport failure from voluntary stopping before attributing a cause.

Mobile Streaming Interruption (MSI)

Definition. Mobile or client-side interruptions can drop an in-progress streaming response, with no recovery of the partial output when the user returns to the session. Screen-lock, backgrounding, and connection drops on mobile clients are common triggers.

Observed pattern. After screen-lock interrupts mobile streaming, the session reopens without surfacing the partial response. The user cannot determine whether generation stopped or completed output was lost in transit.

Operational boundary. Output unavailable after client interruption. Requires client/server evidence; do not infer cause from an empty screen.

Evaluation and proposed mitigation. Correlate lifecycle, network, client, and server traces. Persistent buffering and client reattachment are candidate remedies; a local HTML test cannot establish their effect.

Feedback Persistence Gap (FPG)

Definition. Within-session corrections, scope clarifications, and explicit user preferences do not persist beyond the session unless an explicit memory surface or product mechanism captures them. The same user encounters the same correctable issue in subsequent sessions.

Observed pattern. A user corrects a stylistic or factual error in one session; the same error recurs in subsequent sessions without the correction being applied. Whether corrective feedback is stored or used in later training depends on product policy, consent, and implementation; recurrence alone does not establish the training path.

Operational boundary. Prior correction does not persist across sessions. Only a defect relative to an explicit persistence contract; privacy-preserving non-memory may be intended.

Evaluation and proposed mitigation. Test the stated persistence contract with user consent and visible controls; distinguish memory behavior from later training use.

Retry-Failure Invisibility (RFI)

Definition. An evaluation records the first outcome but omits a collected retry outcome, or reports only a final answer without the first-to-retry transition. This can conceal observed recoveries and regressions. It does not distinguish latent capability from behavior.

Observed pattern. A transcript audit labels an unresolved item as a terminal capability failure while omitting an observed successful follow-up without new user-supplied information. Missing transitions can conceal recovery across several workflow labels.

Operational boundary. Evaluation omits first/second-turn transitions. A measurement blind spot, not a task-level model defect or demonstrated counterfactual.

Evaluation and proposed mitigation. Retain both outcomes and all four transition types. Do not assign a hypothetical successful retry to an untested case.

Stale Third-Party Tool/UI Knowledge (STK)

Definition. Instructions for a third-party interface conflict with its current documented state and match an older version. A mismatch alone is a guidance error; calling it staleness additionally requires dated evidence. The model's actual training exposure is generally unknown.

Observed pattern. The model corrects interface guidance only after the user describes a mismatch in detail. The user bears the cost of recognizing the mismatch and navigating the misdescribed interface.

Operational boundary. Guidance conflicts with current interface. Updated documentation can introduce new task evidence; classify separately from intrinsic text-only correction.

Evaluation and proposed mitigation. Verify procedural claims against dated official sources and the actual interface. A changed interface, unsupported guess, and model knowledge cutoff are different explanations.

Quantitative Inconsistency (QI)

Definition. Within a single response, the model emits two or more numerical claims that contradict each other under the response's own stated assumptions. The contradiction is internally checkable but remains unresolved in the delivered response.

Observed pattern. Breakeven and average-cost figures cannot both hold under the stated unit cost. A later acknowledgment does not establish whether an internal self-audit occurred; the observable error is the unresolved contradiction.

Operational boundary. Numerical claims conflict under explicit shared assumptions. Check units and identities; under-specified tasks are ambiguous, not model failures.

Evaluation and proposed mitigation. Check all linked calculations under common assumptions, including narrative comparisons and boundary cases. A correct final number does not validate its explanation; Study 2 supplies concrete examples.

Phantom Placeholder Persistence (PPG)

Definition. The model issues a command, file path, citation field, configuration value, or code snippet containing an unresolved template placeholder even when the surrounding context implies a concrete value is needed. The placeholder substring is emitted as if it were a final value.

Observed pattern. A command retains a placeholder, a citation retains a template author/year, or code retains a generic path despite available concrete values. Substitution is left to the user and may first be discovered through an execution error.

Operational boundary. Required concrete value remains an unresolved placeholder despite being provided in task context. Do not reject valid angle brackets or shell redirection.

Evaluation and proposed mitigation. Use syntax-aware checks only where the task supplies a required concrete value. Valid angle brackets, shell redirection, and explicitly requested templates are not unresolved placeholders.

Usage-Cap Termination (UCT)

Definition. A rate limit or usage cap terminates the conversation mid-task. The system surfaces a user-facing message but recovery often degrades to a malformed completion rather than graceful state preservation. The continuation turn following the cap event does not reliably resume the task.

Observed pattern. After a usage-cap notice, continuation produces a short completion message without resuming the task. Repeated events within one session compound the disruption.

Operational boundary. Quota or rate-limit interruption. Recovery after access restoration changes system conditions and is outside primary RBF comparisons.

Evaluation and proposed mitigation. Record the limit event and subsequent resume state. Restoration of access changes system conditions; it is not a primary answer-free recovery contrast.

Visual Misreading Cascade (VMC)

Definition. Given a structured visual input (chart, table, dashboard, form, schematic, or culturally conventional diagram), the model produces a confident interpretation. When the user corrects a specific structural element, the model produces a new interpretation that is itself incorrect in a different way. The cascade can repeat across multiple correction cycles.

Observed pattern. User corrections trigger successive confident misreadings of the same visual structure until the user restates it explicitly, without an intervening request to confirm that structure.

Operational boundary. Incorrect structural extraction propagates into interpretation. User-provided structural corrections are informative feedback, not primary answer-free retries.

Evaluation and proposed mitigation. Compare extracted structure with the actual image before evaluating downstream interpretation. User corrections to chart structure add information and belong in a separate feedback condition.

Artifact Finalization Failure (AFF)

Definition. In artifact-producing workflows, the model generates a deliverable (PDF, DOCX, slide deck, spreadsheet, code file) that nominally exists but does not satisfy user-facing readiness criteria. Terminal text quality is acceptable; artifact-level readiness is not.

Observed pattern. A document opens with misaligned headings, broken links, missing sections, or unusable layout; a running code artifact still requires specified formatting repairs. Inspection of the actual file reveals the defect.

Operational boundary. Deliverable violates explicit openability, completeness, structural, or layout checks. Judge the artifact, not the assistant's completion claim.

Evaluation and proposed mitigation. Open and render the deliverable, then check explicit structural and visual requirements. Study 2 shows that page count and extractable text can pass despite an unreadable artifact.

Source Verification Drift (SVD)

Definition. A workflow begins with a requirement to verify sources, claims, citations, or factual assertions, but progressively drifts into plausible synthesis without sufficient source grounding. The drift can be subtle: citations may support adjacent points but not the exact claim, or a source may be stale for a time-sensitive assertion.

Observed pattern. Citations increasingly support adjacent points or stale assertions instead of the exact claims, while their continued presence gives an appearance of grounding.

Operational boundary. A claimed source does not support the associated statement. Check source spans and dates; citation presence is insufficient.

Evaluation and proposed mitigation. Match claims to dated source spans and check whether support changes across the workflow. A citation can be real yet fail to entail its associated claim.

Editorial Boundary Contamination (EBC)

Definition. Process commentary, advisory metadata, or other chat-level material appears inside a deliverable where the original specification excludes it. Required methods, limitations, provenance, and ethical disclosures remain legitimate manuscript content. The artifact-level boundary between conversation and final output fails.

Observed pattern. A draft includes venue advice, a public summary repeats a private chat instruction, or a submitted artifact contains an excluded revision note. Substantive correctness does not prevent this contamination.

Operational boundary. Process/advisory content appears inside a deliverable where prohibited. Legitimate methods, limitations, and disclosures are not contamination.

Evaluation and proposed mitigation. Inspect for prohibited process notes or private chat context. Required scientific methods, limitations, provenance, and AI-assistance disclosures are legitimate manuscript content.

Audience / Context Miscalibration (ACM)

Definition. The model produces technically correct content that is poorly calibrated for the intended audience, register, or distribution channel. The substantive content is acceptable; the communicative fitness for the audience or channel is not.

Observed pattern. A public-communication artifact contains identifying or personal context that should not appear publicly. A summary written for a general audience uses technical terminology better suited to a specialist audience. A short executive update is delivered as a long, citation-heavy academic paragraph.

Operational boundary. Output violates stated audience/channel requirements. Unstated user preferences do not define a failed original task.

Evaluation and proposed mitigation. Evaluate against the stated audience and channel. Unstated preferences do not retroactively define failure; explicit requirements can guide a pre-delivery audience check.

Appendix B: Historical pilots and interpretation

B1 Retrieval pilot

Setup

A fifteen-item heterogeneous retrieval task. Items belong to three structural classes: (1) direct URLs with human-readable paths (n=9); (2) URLs containing opaque numeric identifiers (n=3); (3) title-only references requiring search (n=3).

The historical report records twelve initial resolutions and fifteen resolved items after "try a different approach" in a selected fifteen-item task. The three reported recoveries involved opaque numeric URL identifiers; nine human-readable URLs and three title-only items were initially resolved. The private transcript has not been independently reproduced. The initially successful items may have been carried forward rather than independently re-answered, so the recorded workflow state cannot establish absence of regression under a fully branched retry design.

Table 18. Historical workflow-resolution scope. Historical aggregate outcomes. Intervals describe binomial calculations, not a justified population sampling model for this selected single-batch task.

Quantity	Count	Estimate	Wilson 95% interval
Initial resolution	12/15	80.0%	54.8-93.0%
Resolved after workflow retry	15/15	100.0%	79.6-100.0%
Joint recovery incidence	3/15	20.0%	7.0-45.2%
Recovery among initial failures	3/3	100.0%	43.9-100.0%

Paired and exploratory calculations

Conditional on treating the fifteen recorded before/after states as matched binary pairs, the exact two-sided McNemar calculation uses three discordances in one direction and zero in the other:

$$p = 2 \Pr\{\text{Binomial}(3, 1/2) = 0\} = 0.25.$$

The twenty-percentage-point change is descriptive and does not reach the conventional five-percent threshold. The calculation is not a remedy for within-batch dependence, selective item inclusion, or incomplete retesting. We report a Wilson interval for the joint recovery proportion, not as a general confidence interval for a paired net difference with unknown regression probability.

The post-hoc opaque/non-opaque association table is $[[0, 3], [12, 0]]$ (success, failure columns). Its two-sided Fisher value is 0.002198. This is the minimum attainable value under those observed margins. Recoding one opaque item as initially successful gives $p=0.028571$; recoding two gives $p=0.20$. These are descriptive sensitivity calculations for that table. They do not establish a general trigger effect, protect against event-selection bias, or demonstrate robustness to arbitrary coding errors. The association is omitted from the abstract's evidential lead.

Supported conclusion

The report documents three recoveries in one selected workflow, with no independently reproduced transcript audit. It motivates controlled measurement of retrieval escalation. It does not establish the prevalence of PTS, vendor differences, an opaque-identifier law, or a mechanism distinct from additional search effort and inference computation.

B2 Historical model and retry pilots

The v1.23 manuscript reported a ten-task, three-model pilot and a five-task, four-condition pilot using named Anthropic configurations. Under the historical lexical rubric, each reported model passed both QI, PPG, EBC, and ACM items and failed both RETRY_PROXY items. In the four-condition aggregates, scores sometimes fell after retry: for example, Sonnet moved from 4/5 initially to 2/5 after "Please continue," and Haiku from 3/5 initially to 2/5 after verification. These are outputs of historical rubrics, not

independently validated model-performance estimates. Full cell tables, figures, and calculations are preserved in the historical supplement.

Audit of historical execution and scoring

The inspected public snapshot, commit 5ae5f6a3dd6f (<https://github.com/vjgits/Research-Papers/commit/5ae5f6a3dd6f7485dbbb479f56c4fdf7d1814ee6>), contains historical raw/scored material for the latter but not the raw results claimed for the former. The released v0.2 runner was a mock scaffold, did not implement the described provider execution and shared-anchor branching, and included syntax errors. These are reproducibility limitations, not evidence that the historical calls did not occur.

More importantly, several acceptance tests did not measure task success. The quantitative task confused a stated cost with revenue. The placeholder scorer rejected shell redirection syntax. The abstract task omitted the topic and findings yet could pass on an isolated word. Audience and retrieval checks substituted vocabulary for semantic compliance; retrieval lacked an actual retrieval environment. “Please continue” invited an incremental answer but was scored as a standalone replacement.

Accordingly, historical pass-rate changes are retained only in Appendix B as outputs of those rubrics. No ranking or substantive model regression claim is made. Neither an LLM judge nor semantic similarity is presumed to fix the problem without validation. A corrected scorer cannot reconstruct missing raw responses or missing request provenance. The successor materials provide new acceptance tests and explicit branching; they do not retroactively validate the old experiments.

The reported counts remain part of the historical record. They are not pooled with either development study, and a corrected scorer cannot reconstruct missing request provenance or absent raw answers. Their scientific role is to document why execution traceability and task-valid acceptance criteria are necessary before interpreting a pass-rate change.

Appendix C: Additional reported observations

Four additional author-reported workflow observations are summarized below. They are hypothesis-generating cases; their inclusion does not establish independent replication or recovery under a fixed, non-answer-bearing retry. The dated source memo is identified in the companion provenance record.

Table 19. Additional author-reported observations and their evidential treatment.

Memo ID	Reported pattern	Evidential treatment
RBF-2609-01	Table/link output did not fit a viewport; plain lines after a reminder	Candidate AFF/ACM; rendering success depends on client state and needs direct verification. Constraint reminder is procedural feedback.
RBF-2609-02	References supplied without fetch checks; later link audit	Candidate SVD; memo reports five problematic links, but a redirect alone is not necessarily a dead source. Fetching adds evidence from existing tools.
RBF-2609-03	Broad inability claim narrowed after a contradiction was pointed out	Candidate capability-description error. Model self-report does not establish cause; diagnostic user feedback makes it distinct from fixed generic retry.
RBF-2609-04	PDF editing initially described as impossible; later reframed as reconstruction risk	Same candidate family within the same session. Inspection can establish feasible reconstruction, not identical source recovery or error-free editing.

These reports do not add quantitative events to either study. Repeated observations in one session are not independent cross-session replication. Diagnostic feedback, changed access, and newly supplied information must remain distinguishable from a fixed procedural retry.

Appendix D: Review provenance and amendments

A subsequent AI reviewer scored an enriched 75-output packet containing all 35 failed-or-uncertain outputs and 40 masked passing controls. It had prior exposure to manuscript findings and principal error counts. Although packet model and condition identifiers were withheld, this was not independent blinded validation; wording could reveal verification. Its self-reported model identifier was not authenticated. No new subject responses were collected.

The review record distinguishes claimed scope from performed checks. The reviewer later disclosed that its original v1.29 manuscript reading covered pages 15–32, not pages 1–14. The original arithmetic review is therefore scoped to Appendix B; the separately executed packet arithmetic is also retained. Broader Study 1 and Study 2 reconciliations were reported only after the scope correction. The reviewer also withdrew claims that Section 2 formalization and configuration disclosure were new in v1.30: both were already present in v1.29. Later confirmation does not retrospectively validate an earlier verification claim. These corrections limit the weight of the AI review and do not alter the frozen experimental scores.

The initial submission assigned 38 overall passes, 35 failures, and two uncertainties. Three visual responses had uncertain content because the reviewer could not access the comparison image. The image was present in the original ZIP and matched its recorded checksum; direct resupply resolved the access problem. A separate amendment changed those three content judgments to pass: two overall uncertainties became passes, while one response still failed format. The combined review therefore assigns 40 overall passes and 35 failures, 66 content passes and nine content failures, 46 format passes and 29 format failures, and 66 target-negative versus nine target-positive judgments. Original review, amendment, frozen screening, and human marks remain separate.

Equal overall totals conceal twelve content/target differences from frozen screening. The reviewer changed eight BOUND uncertainties to pass/no by evaluating the final notice separately from prefatory process commentary; it changed four HTML uncertainties to pass/no using document inspection. The BOUND rule refers to production material entering the notice, leaving the boundary between the final notice and the complete response insufficiently explicit. AUD instead has an explicit prohibited-word rule applied to the whole response. These task-specific scopes require a clear definition, not silent harmonization after scoring. Future adjudication should record whole-response format, final-deliverable content, and commentary contamination separately. Alternate scope readings belong in a labeled sensitivity analysis; they do not replace frozen uncertainties.

The visual amendment also raised a possible panel mismatch. The three affected cases correctly use the single-panel comparison input; the scenario input separately contains Planned, Actual, and Previous week panels. This distinction intentionally varies the selection burden. No attachment substitution or frozen task correction is warranted. Reviewer comments also identify possible self-endorsement cues: some answers announce their correctness before presenting the deliverable. Audit instructions should direct raters to task evidence and preserve full responses rather than allowing those assertions to substitute for checking.

The same chronology applies to artifact checking. The reviewer initially hand-transcribed four distinct HTML documents, stripping commentary from four saved responses, but described its coverage too broadly. Its correction supplied programmatic verbatim and extracted-document renders for nine cases,

with a log and one screenshot; the original browser version and hand-transcribed files were not supplied. Logged answer-hash prefixes matched the archived responses. The subsequent author-coordinated rerun in Section 3 records its own browser version, full hashes, and all eighteen screenshots. It establishes a new reproducible check, not the execution history of either earlier procedure.

D1. v1.34 scope rescore added after external review

An external reviewer argued that the 29-regression headline was scoring-scope dependent because the E instruction asks the model to check the prior response before returning a corrected answer. The reviewer recommended rescoring all E outputs under a final-deliverable-only scope. This criticism was accepted.

The resulting v1.34 analysis preserves the frozen whole-response labels and adds a second deterministic scope across all 576 responses. Fixed-line tasks use the final N non-empty lines specified by the original task; HTML uses the document from the first doctype marker. Excluded prefatory text is separately coded as boundary contamination. Under this rule, all 29 original E failures pass the final-deliverable checks, while the same 29 remain contamination events. A-to-E transitions change from 2 recoveries / 29 regressions under whole-response scope to 2 recoveries / 0 regressions under final-deliverable scope.

Because the rescore was added after collection and after the outputs had been inspected, it is explicitly labeled post hoc. The archive includes the rule, code, per-response comparison, and hashes so the result can be reproduced without a model call. No historical human mark or frozen screening label is overwritten.

D2. v1.36 HTML suffix check and release corrections

A subsequent external audit mutation-tested the parser used through v1.35 by appending a trailing commentary line to every response that had passed final-deliverable scoring. All 506 fixed-line passes then failed their task-specific extraction/checks, but all 64 HTML passes remained accepted because the HTML parser tolerated text after the closing `</html>` tag and the contamination flag inspected only text before the document start.

Version 1.36 therefore adds a post hoc HTML suffix-contamination check: the final deliverable is bounded from the first HTML document-start marker through the final `</html>`, while non-whitespace text before or after that range is recorded as boundary contamination. Direct inspection found no trailing text in any of the 64 saved HTML responses, so this addition changes no observed counts. The archive includes a mutation test demonstrating the fixed-line behavior and the new HTML suffix flag.

The same release audit identified three packaging/rendering defects: the independent-rater answer key was shipped beside the packet, Unicode superscript characters produced inconsistent PDF weight, and inline-code backticks around archive paths rendered literally. Version 1.36 withholds the mapping while publishing its SHA-256 commitment, uses real superscript formatting for the exponent, and removes the literal backticks. A later transport check showed that a separately transmitted standalone PDF can acquire additional provenance metadata even when its visible text is unchanged. Accordingly, the archive manifest is the authority for the manuscript PDF hash; a standalone copy intended for deposit should match that manifest value after download.

D3. Secondary sensitivity retained for continuity (Study 1)

Before the dual-scope rescore, a post hoc analysis excluded AUD, BOUND, and HTML, where lexical scope, production commentary, or browser behavior complicated interpretation. Under whole-response scoring, the remaining six families contained 96 A/E pairs with two recoveries and eleven regressions, for

a -9.4 percentage-point net change. This shows that the original whole-response effect was not confined to the three most disputed families.

The exclusion is now secondary because the archive-wide dual-scope analysis addresses the underlying ambiguity directly. It remains a post hoc sensitivity description, not independent confirmation. Neither sensitivity analysis replaces the frozen original labels.

Appendix E: Study H disagreements and unsure judgments

The entries below retain every reported disagreement and unsure judgment. The accompanying release preserves the analyzer output verbatim, including its original position-denominator label; Table H1 explains that denominator. H11 remains in the reported calculations despite its exclusion from substantive probe interpretation.

Table H3. Study H case-level disagreements and uncertainties

Case	Stratum	Reported judgment or disagreement
H02	A (real)	Rater 1: no; rater 2: unsure.
H04	A (real)	Rater 1: unsure; rater 2: no.
H11	B (probes)	Rater 1: yes, both; rule: no, none. Rater 2: no. Defective completeness probe.
H16	B (probes)	Rater 1: yes; rater 2: unsure. Commentary only, no deliverable.
H21	B (probes)	Both raters: after; rule: before. Trailing commentary sentence.
H25	B (probes)	Both raters: both; rule: before. Fenced fixed-line payload.
H39	B (probes)	Both raters: after; rule: before. Trailing one-word sign-off.
H40	B (probes)	Rater 1: yes; rater 2: unsure. Trailing note after Markdown table.

Appendix F: Evidence provenance and observation scope

F1. Evidence tiers

The evidence tiers distinguish reported observations from tested explanations:

- Tier 1: Reported aggregate pilot outcomes; the historical transcript is not independently reproduced. Study 1 separately supplies archived response-level records.
- Tier 2: Reported trace-backed observations; independent trace verification is not claimed here.
- Tier 3: Abstracted cross-platform workflow summaries (artifact and editorial-boundary extensions).
- Tier 4: Hypothesized mechanisms requiring controlled replication.

F2. Coding procedure

Each event records task context, original output, user retry, post-correction outcome, new substantive information, failure mode, trigger, recovery type, and user-visible cost. Recovery requiring new user information is excluded from RBFs.

F3. Responsible data handling

The archive separates Study 1 inputs and de-identified human labels from Study 2 context, reconstructions, responses, and artifacts. Historical private transcripts, account identifiers, and full personal exports are excluded. Task IDs, stage labels, input/output hashes, and dated amendments retain provenance.

F4. Provenance and limitations of the case material

The v1.23 manuscript describes fourteen naturalistic sessions from one researcher and abstracted cross-platform workflows, without a complete sampling frame or event census. Raw transcripts were not released, per-message model identity was inconsistently recorded, and attribution partly relied on recall. Study 2 inspects selected exports, the source paper, and reconstructed inputs; it does not independently revalidate the historical bank.

The tiers separate provenance from interpretation. Earlier “transcript-verifiable” claims do not mean independently transcript-verified. The bank can generate testable tasks but cannot estimate frequencies across users or vendors. Independent event counts by mode, overlap rates, and inter-annotator agreement are unavailable.

F5. Hierarchy and coding rule

The sixteen original labels are retained for continuity, not as mutually exclusive model defects. Five workflow families organize eligible observations: retrieval termination/escalation (PTS, SMI), batch allocation (BDTF), verification omission by modality (QI, SVD, VMC), artifact finalization (PPG, AFF), and boundary/audience control (EBC, ACM). System/product events (MRT, MSI, UCT, FPG), knowledge-update conditions (STK), and the evaluation blind spot (RFI) form separate layers.

The unit is one task attempt and its bounded continuation. Annotators score success and eligibility, then assign the earliest evidenced failed requirement as the primary workflow family. Secondary labels are allowed; causal locus remains “unknown” without trace or telemetry support. Held-out evidence and reliability may justify merging or splitting labels. Appendix A gives definitions and exclusions.

Appendix C adds four author-reported observations as hypotheses, not a seventeenth validated mode or independent recurrence count. Pointing out a contradiction supplies diagnostic feedback even without the final answer and differs from a fixed generic retry.